AZURE MASTER ARCHITECT PROGRAM · AUGUST 2026

Unified Digital Citizen Services Platform.

One front door for 2.1 M Nordic citizens across DK · SE · NO

ARCHITECTURE, AI & AGENTIC SUBMISSION

2.1 M	12	47 → 1	28 → 4 d
citizens · DK · SE · NO	languages	portals unified	processing time

What this is, and what it isn't.

🎯 A target vision
The production-target architecture for a federated Nordic platform, end to end.

⚙️ A working core
Deployed on a real Azure tenant with one command; the live demo runs on it.

📐 Some blueprint
A few bricks are designed and registered, not yet switched on, each with a roadmap gate.

Every claim in this deck carries one honesty label:

- Live: on the tenant today
- Implemented: merged & smoke-tested
- Scripted: installer phase, idempotent
- Blueprint: designed, not yet live
- Roadmap: external dependency

> UDCSP is a production accelerator: a target vision, a working core you can run today, and a named path for the rest.

The problem.

2.1 M

citizens across DK SE NO

47

separate legacy portals

28 d

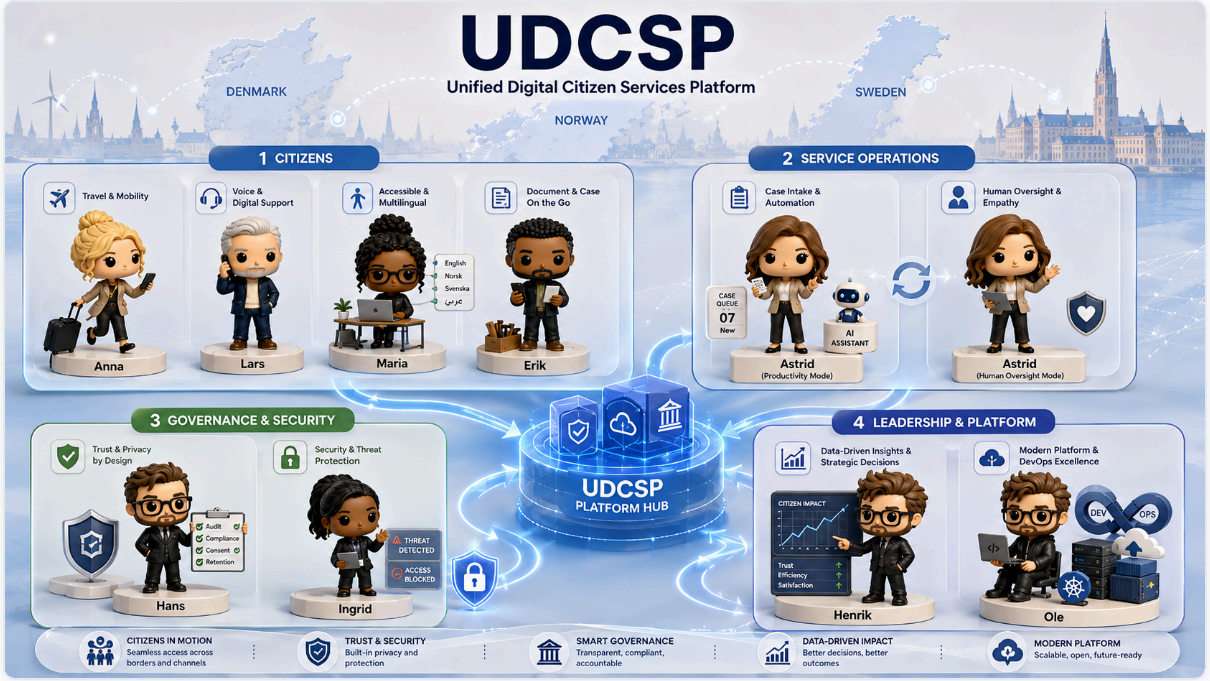
average time for a residency case

A citizen who moves from Copenhagen to Stockholm proves their identity again, re-sends the same papers, waits 28 days for a decision, and uses a portal that may not speak their language, or may not be accessible at all.

UDCSP unifies the identity and the experience, then connects to the national systems to share and check information, never to replace them.

GDPR · EU AI Act · eIDAS 2.0 · NIS2 · WCAG 2.1 AA · ePrivacy

Who we build for.



CITIZENS

Anna · Lars · Maria · Erik

Anna moves country. Lars is blind and calls in. Maria uses a screen reader. Erik uploads from his phone.

OPERATIONS

Astrid, caseworker

The AI helps her work faster. She always makes the final decision.

GOVERNANCE

Hans · Ingrid

Hans protects privacy and consent. Ingrid watches for security threats.

LEADERSHIP

Henrik · Ole

Henrik needs clear data to decide. Ole runs the platform with one command.

The insight: unify the identity, connect the systems.

We did not rebuild the national portals. We unified the one thing that was missing: a single identity and a single experience, and connected it to the systems that already hold the records.

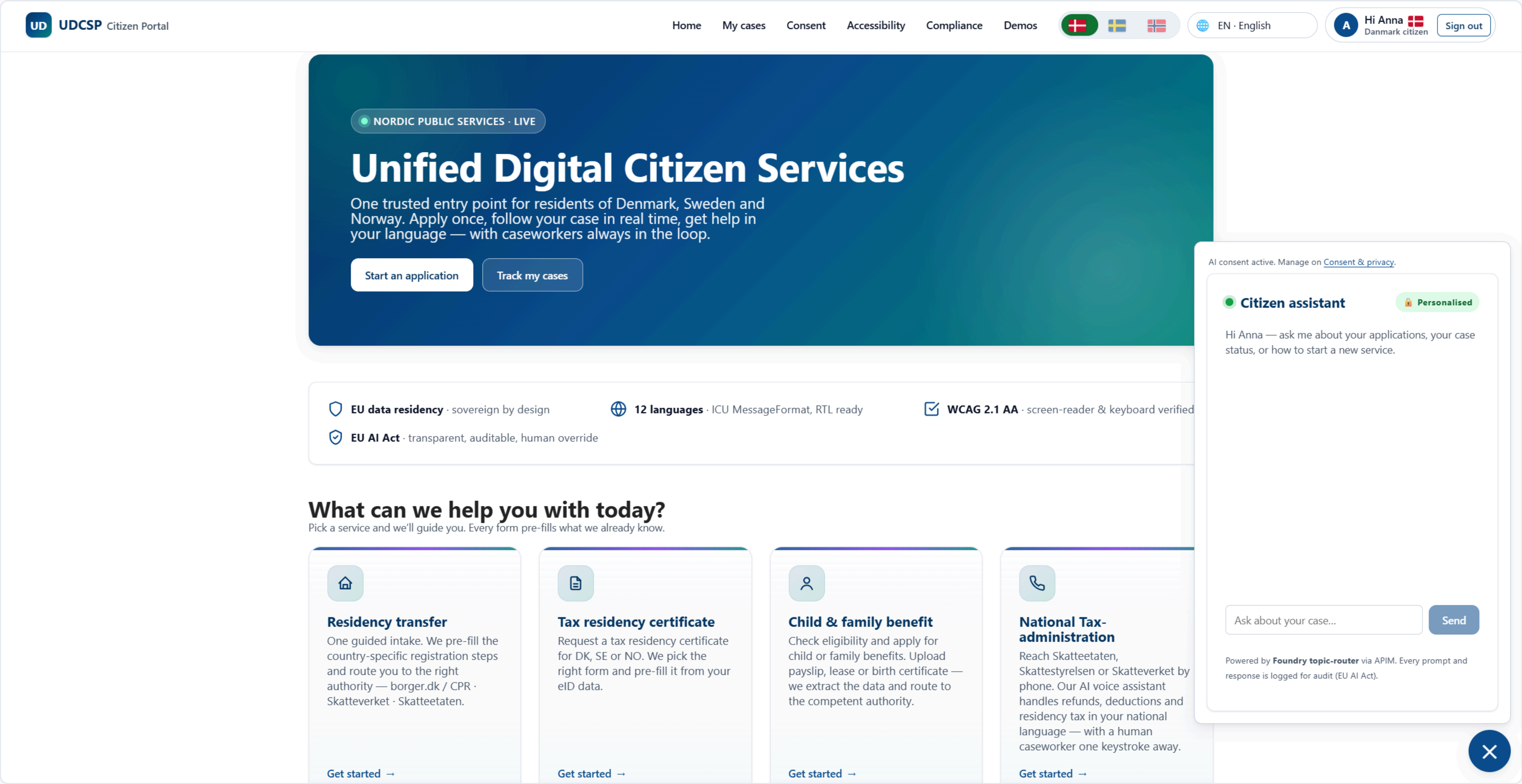
The citizen signs in once with their national eID. From there the platform shares and checks information with each authority, pre-filling the form, submitting it, following the status live. borger.dk, Skatteverket, NAV, Altinn and UDI stay the system of record and the decision-maker; we connect, we never replace.

> Sign in once. Share and check, don't re-collect. Follow the case live.

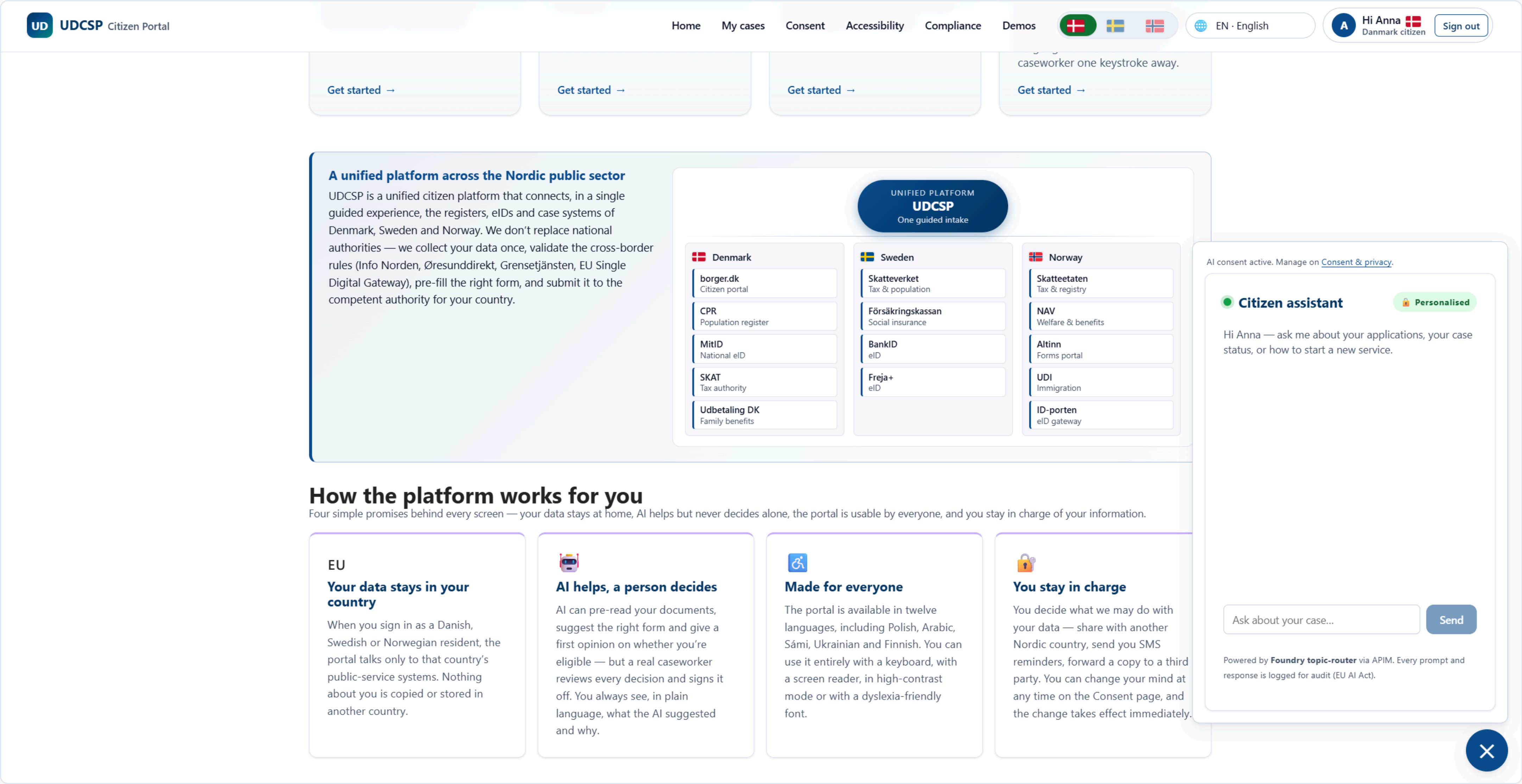
WHY IDENTITY-FIRST, NOT ANOTHER PORTAL

- One identity, one design, one accessibility standard
- Information is shared and verified, not re-collected
- The same code on web, mobile and phone
- A new service is a new card on the same shelf, not a new website
- The authorities keep their data; we keep the experience

One front door: what the citizen lands on.



One identity, connected to every national system.

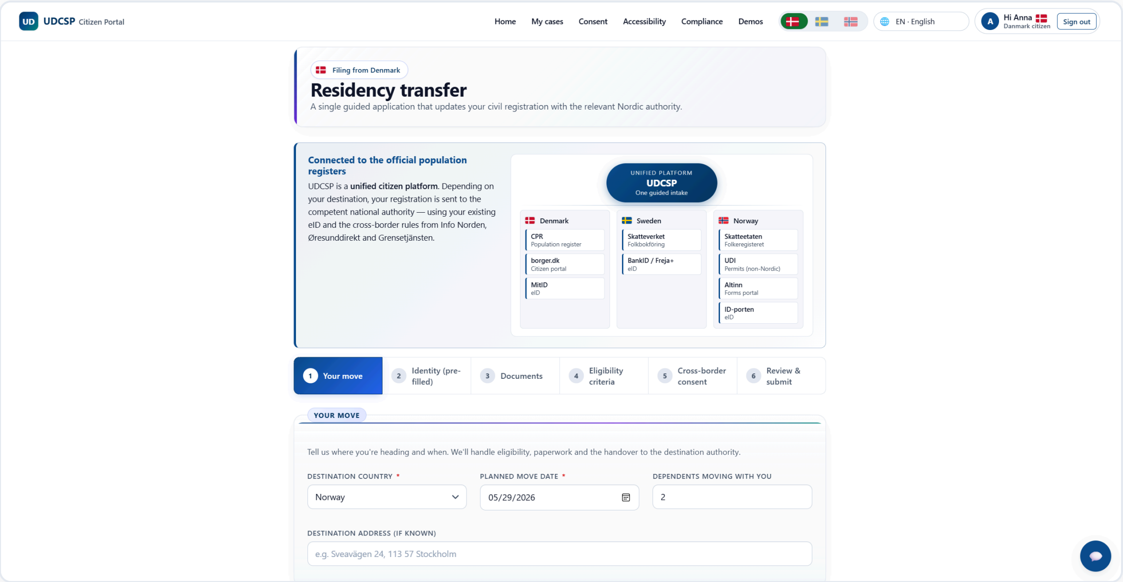


How the front door routes.

One guided intake. The portal finds which authority owns the request and sends it there, using each citizen's national eID and the public cross-border rules.

- **Denmark**: borger.dk · MitID
- **Sweden**: Skatteverket · BankID / Freja+
- **Norway**: UDI · Altinn · ID-porten

The rules come from Info Norden, Øresunddirekt and the EU Single Digital Gateway, following public policy, not logic we invented.



What we ship.

CHANNEL

🌐 **Web · Mobile · Voice**

One web app · 12 languages · accessibility-tested · one free phone number per country

AI BRAIN

🧠 **One brain · 7 agents**

One orchestrator sends work to six specialists on Azure AI Foundry

OUTCOME

📉 **28 days → 4**

Cross-border residency in four days · 47 portals become one · 12 languages

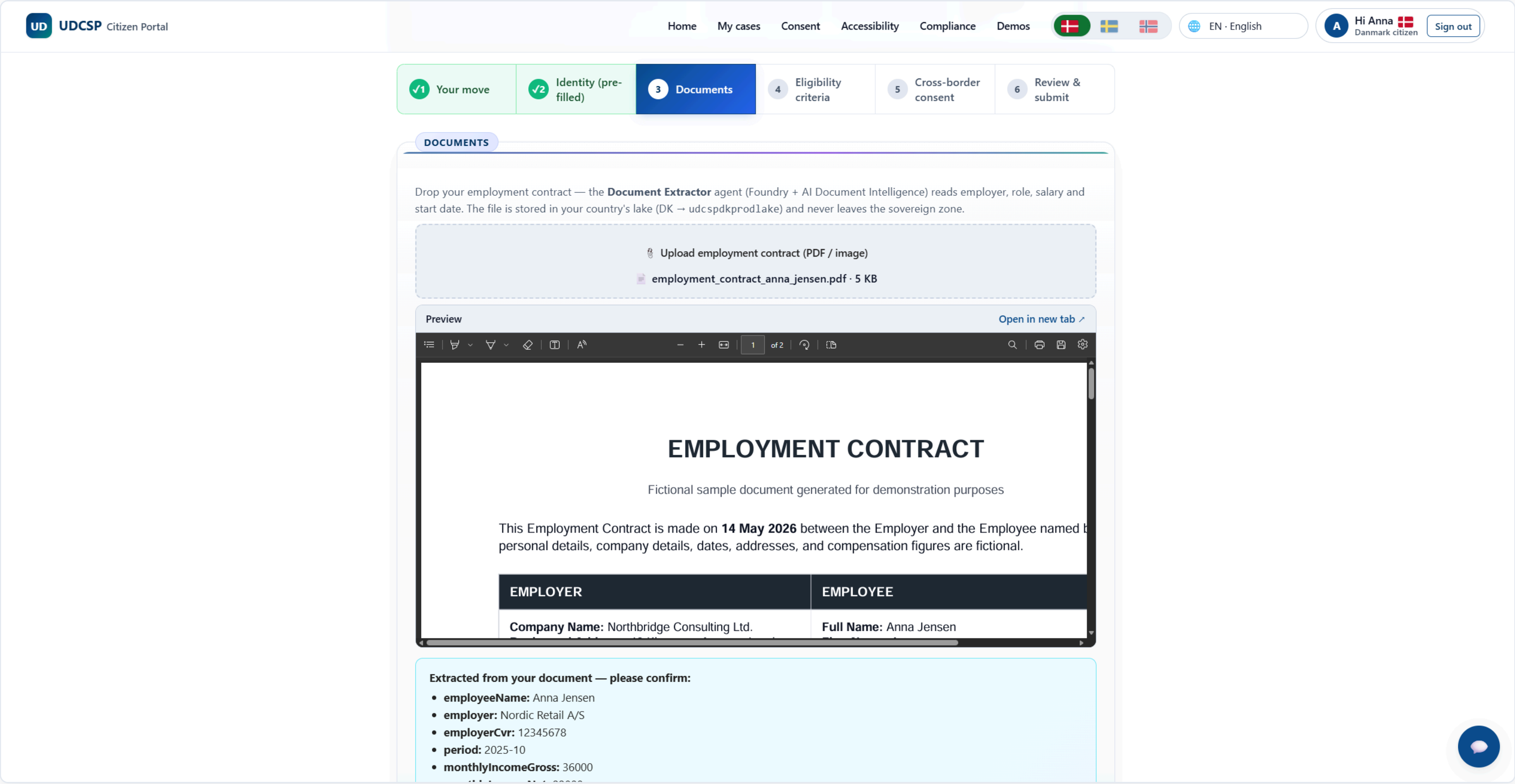
SOVEREIGN

🛡️ **3 zones**

One AI hub and one data zone per country · no data crosses a border uninvited

> UDCSP is a unified citizen platform, not a replacement. The national authorities still make the decision.

Sign in once: the rest is pre-filled.



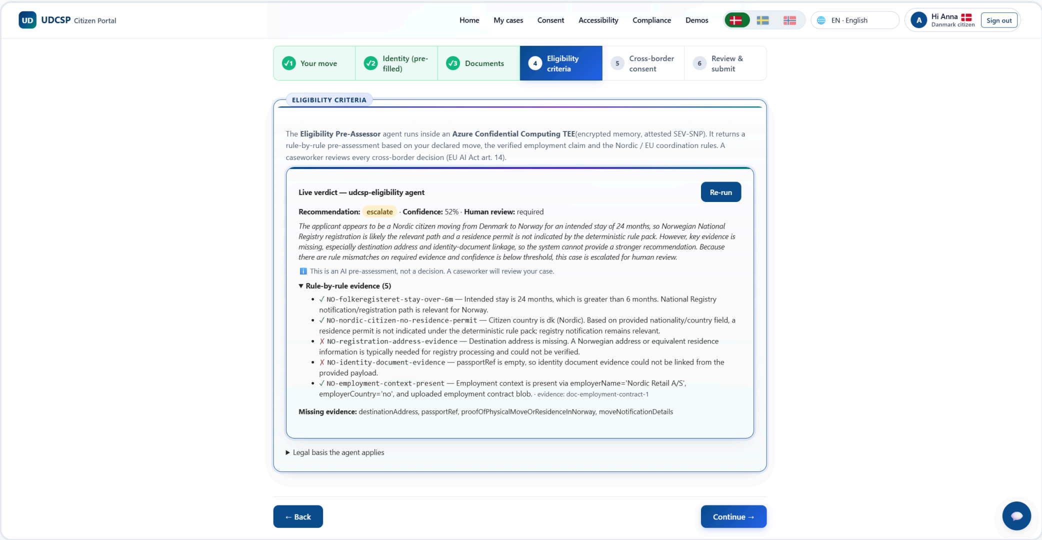
The platform you actually see.

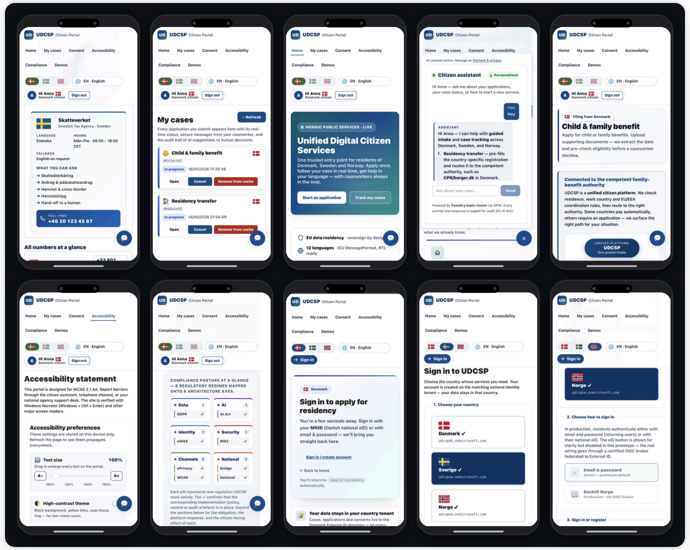
One web app · 12 languages · 3 channels · always accessible.

The same React and TypeScript code runs on phone, tablet and laptop, and adapts to each screen.

The chat assistant stays in the corner. The accessibility menu offers slow speech, high contrast and reduced motion.

On every *Apply* page, the citizen sees the AI result (confidence, evidence, missing documents) before they consent.

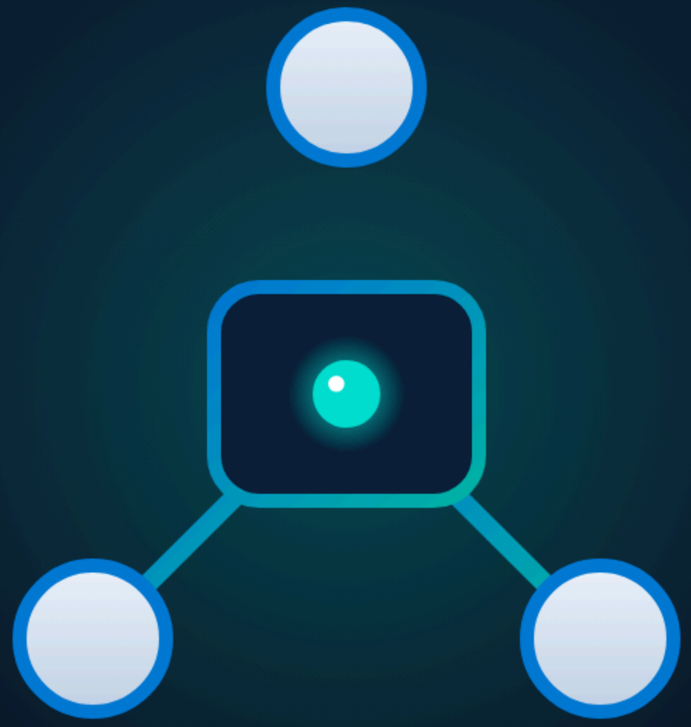




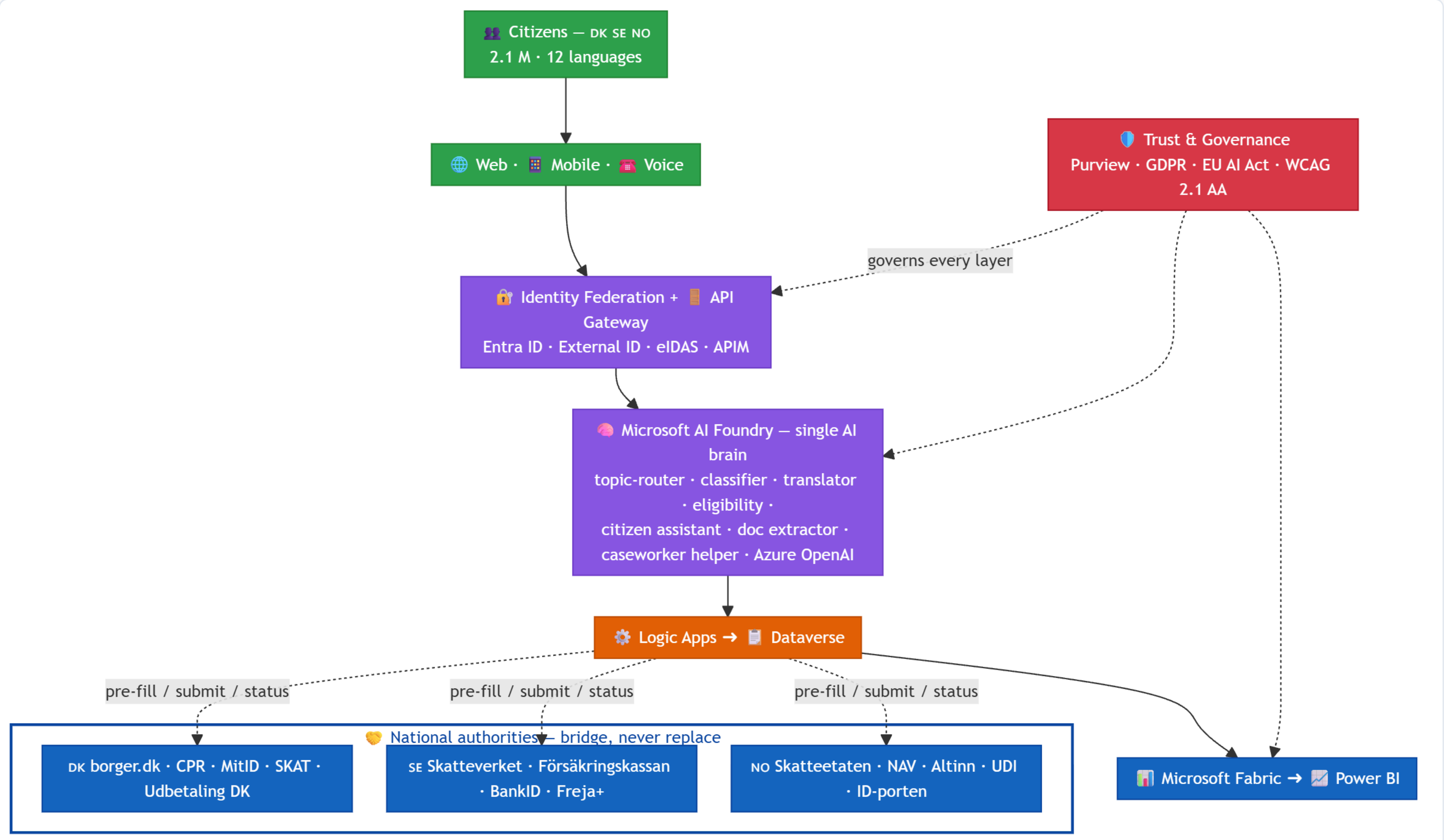
Architecture.

How we built it: three sovereign zones, one private path from the portal to the case

01



High-level architecture.



One country, one hub: three times.

A HUB PER COUNTRY

Each country has its own Azure region, private network, identity system, and AI hub on Azure AI Foundry.

- 🇩🇰 **Denmark**: North Europe region
- 🇸🇪 **Sweden**: Sweden Central region
- 🇳🇴 **Norway**: Norway East region

No AI call ever crosses a border.

THE RIGHT MODEL FOR EACH JOB

- **Real-time speech** model: the voice channel
- **Strong reasoning** model: the hard decisions
- **Fast** model: simple routing and sorting

The model name comes from one parameter file, so a region change is a one-line edit.

> **One honest exception.** Real-time speech is not available yet in Norway, so Norwegian calls use the Sweden hub, inside the EU Data Boundary. The audio still stays in Norway. When the model arrives, one parameter brings it home.

Privatising every zone: from the portal to the case.

One public door per country. Everything behind it is private, from the citizen's first click to the caseworker's queue.

01 **One public front door**

Azure Front Door and a web firewall. Nothing else is public.

02 **A private API gateway**

Azure API Management checks the token, the only way in.

03 **Backends on private endpoints**

Workflows, agents, storage, database, vault: off the internet. Private DNS per country.

04 **The case lands in Dataverse**

One case environment per country, in-region; Power Apps over Dataverse today, D365 Customer Service when licences land. Writes go through workflows, reads through the gateway.

05 **Analytics copies only numbers**

Dataverse sends metrics to Microsoft Fabric. The raw rows stay in the country.

THE ONLY OTHER PUBLIC ADDRESS

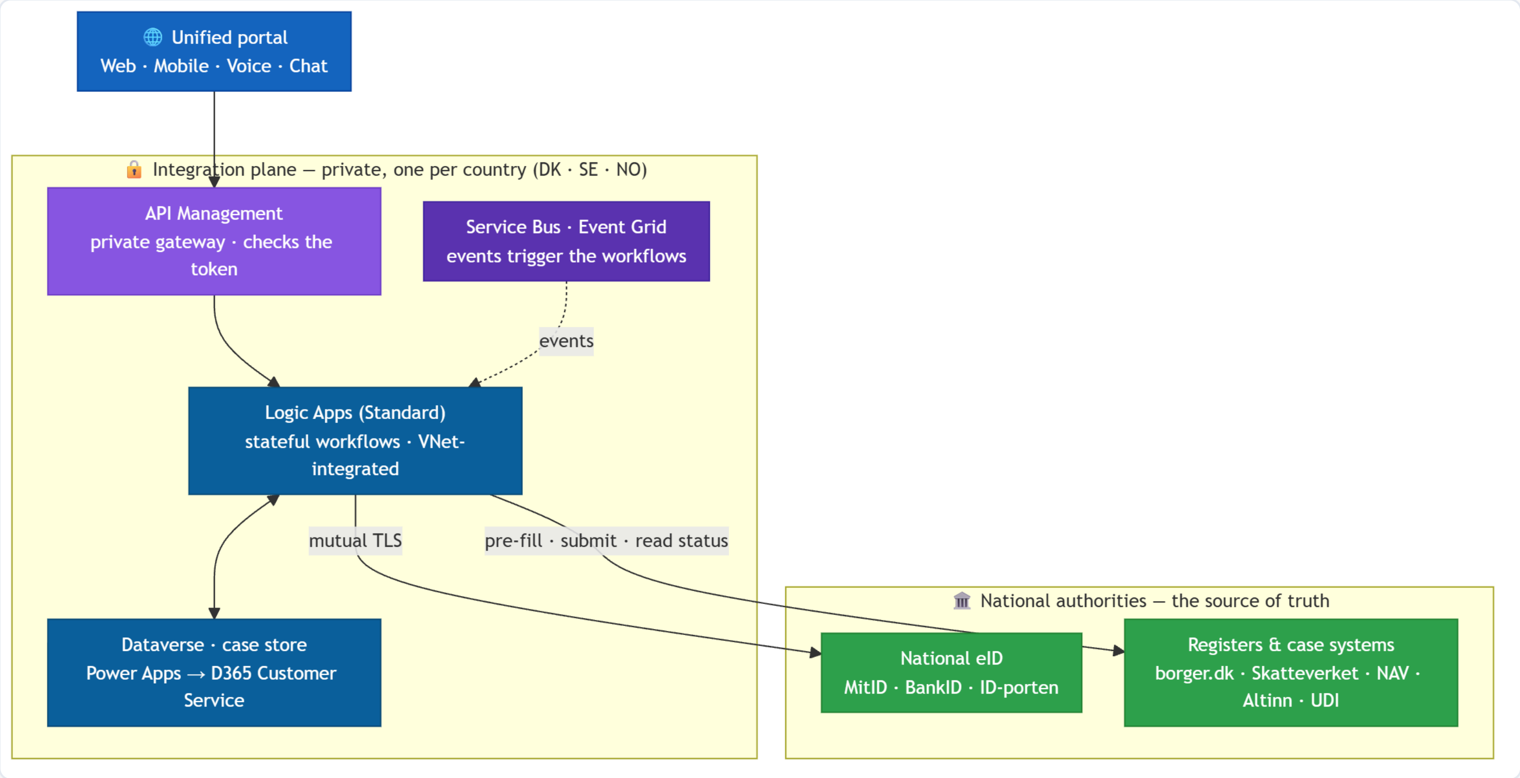
Azure Bastion is the single admin door per country. All outbound traffic goes through Azure Firewall, with allow-lists and TLS inspection.

ENCRYPTED EVERYWHERE

- Customer-managed keys, one set per country
- Two-way TLS to the national authorities
- Field-level encryption for national ID numbers

The integration bridge.

One private integration plane per country. The portal never calls a ministry directly; Logic Apps do, on its behalf.

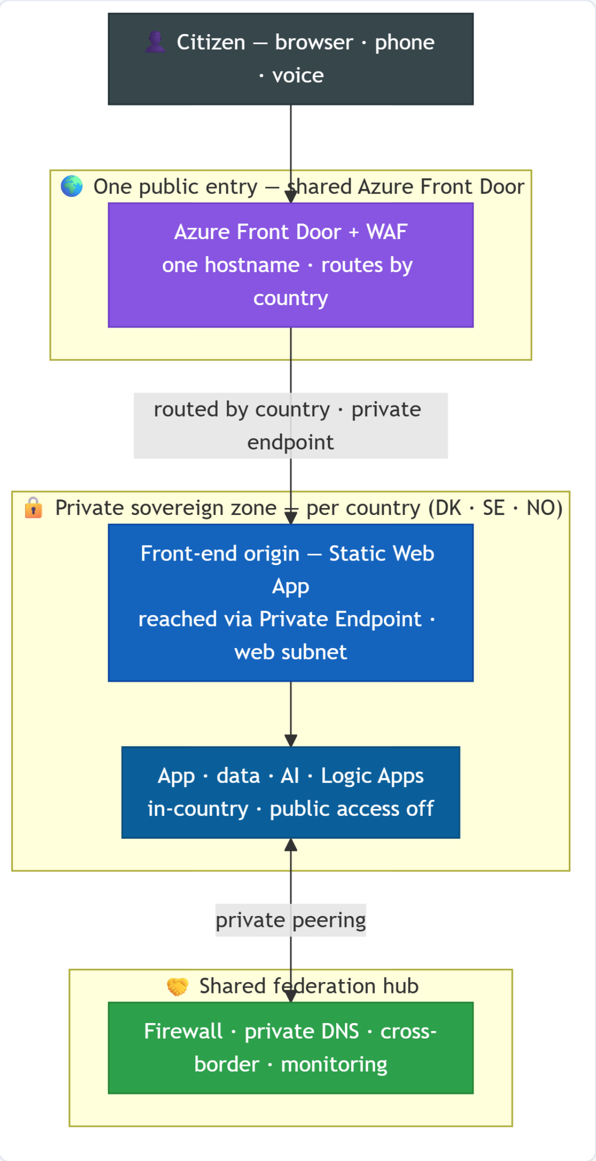


Logic Apps are the bridge: stateful workflows that pre-fill the national form, submit it, and read the status back, writing each case into Dataverse and reaching authorities over mutual TLS. In production they're VNet-integrated.

How the network fits together, simply.

One entry, one experience: private and sovereign underneath.

- **One public entry**: a single Azure Front Door (web firewall) and one UI codebase; the citizen always sees the same portal.
- **Sovereign by design**: each country has its *own* private front-end origin : a Static Web App reached through a private endpoint in that country's `web` subnet, with no public origin.
- That `web` subnet in each spoke is the private door to the front-end, not a separate copy of the app.
- App, data, AI, Logic Apps are private and in-country too; a shared hub handles firewall, DNS, cross-border and monitoring.



Design patterns we use.

Agents-as-Tools
On a phone call, the speech model decides by itself when to call a function or pass to a human.

Gateway + Saga
One API gateway per country. The cross-border case is a 6-step workflow that rolls back if a step fails.

System of record: Dataverse
Dataverse holds each case, surfaced via Power Apps today, D365 Customer Service tomorrow. Writes through workflows, reads through the gateway with a cache.

Strangler fig
The caseworker app writes to a generic Dataverse table today, the canonical D365 Customer Service case entity tomorrow with one switch.

Sidecar
The voice service pairs a call handler with a real-time audio bridge.

Circuit breaker
A failing national endpoint switches to a human queue; citizens never see a timeout.

Scaling to thousands of services.

01 **Orchestration lives in one place**

Per country, the Logic Apps integration plane behind the single gateway, never from the browser.

02 **A journey is a saga**

It fans out to each service in parallel, waits for all, and rolls back cleanly if one step fails.

03 **A new service is declarative**

A workflow + one gateway operation + one catalog entry, no new infrastructure.

04 **Still findable**

One front door, one search → the Topic Router routes to the right service, on web · mobile · voice, 12 languages.

ONE VISA REQUEST → MANY SERVICES

- **Identity**: civil registry + national eID
- **Housing**: municipal address registry
- **Income**: employer + benefits data
- **Tax**: national tax authority
- **Vehicle**: vehicle registration
- **Healthcare**: health enrolment

> Add the 200th service the same way as the 2nd: one workflow, one gateway operation, one catalog entry. The platform grows by configuration, not rewrites.

Meeting old systems where they are.

Anti-corruption layer
One thin adapter per backend. The mess of each old system stays at the edge; the core model stays clean.

Identity resolution
Phone number · national ID · case number, each mapped to one canonical citizen. The backend never adapts to us.

Two-way transformation
REST, SOAP or flat file → our common shape and back. Request transform at the gateway, mapping in the workflow, mutual TLS to partners.

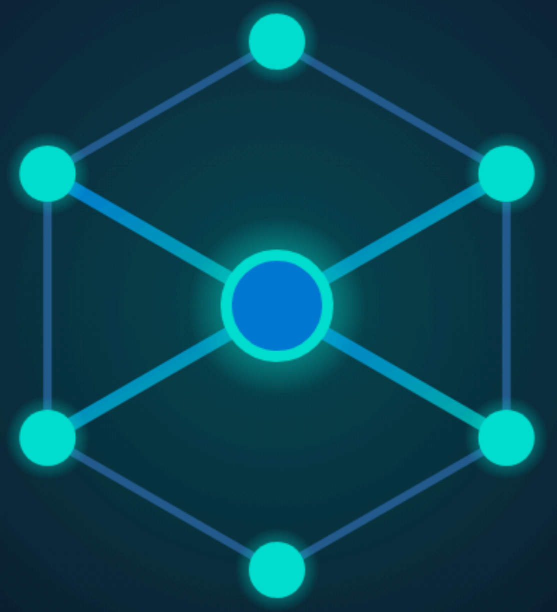
Once-only + catalog
EU Once-Only evidence exchange means we never ask twice. Purview catalogs every field and its lineage.

> We adapt to the backend; the backend never adapts to us. A new connector is configuration and an adapter, not a platform change.

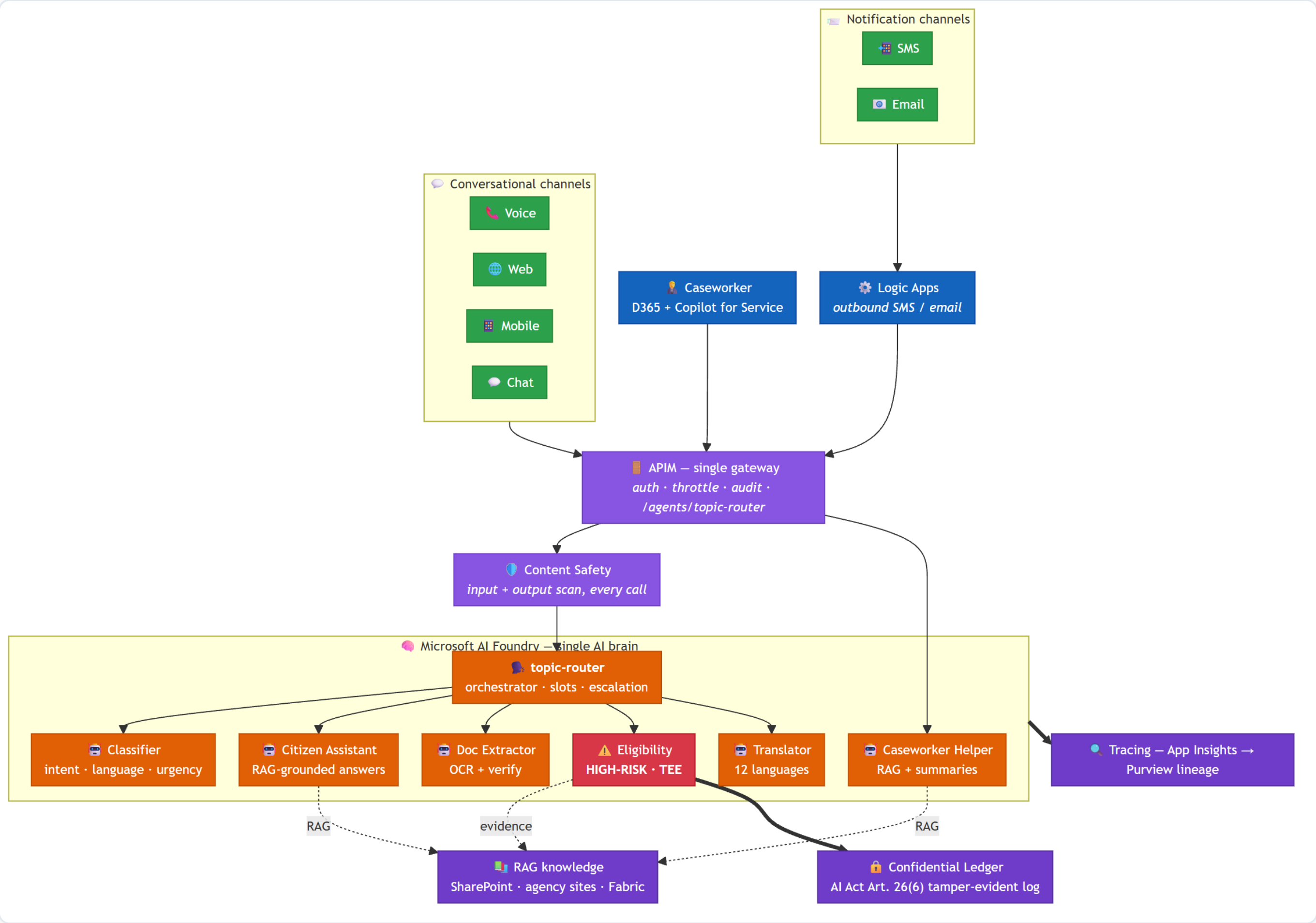
The AI Brain.

One brain on Azure AI Foundry · one orchestrator · six specialists · safe by design

02



The AI Brain: every channel, one gateway.



One brain, not seven chatbots: every model call goes through Azure AI Foundry, routed by the Topic Router.

How a citizen's question flows through the brain.

01 **The channel asks the brain**

Web, mobile and phone reach the Topic Router through the one private gateway, never a model directly.

02 **The router understands and remembers**

It finds the intent and the language across 12 languages, and keeps the short conversation for 24 hours.

03 **It passes the task to a specialist**

Classifier, Translator, Document Reader, Citizen Assistant, Caseworker Helper, or the high-risk Eligibility agent.

04 **The specialist does one job well**

The Assistant answers only from cited sources; the Reader extracts a payslip; Eligibility proposes a result with evidence.

05 **The answer returns, traced and sourced**

Every reply carries a trace number and its sources. On voice, the model decides when to call a tool or a human.

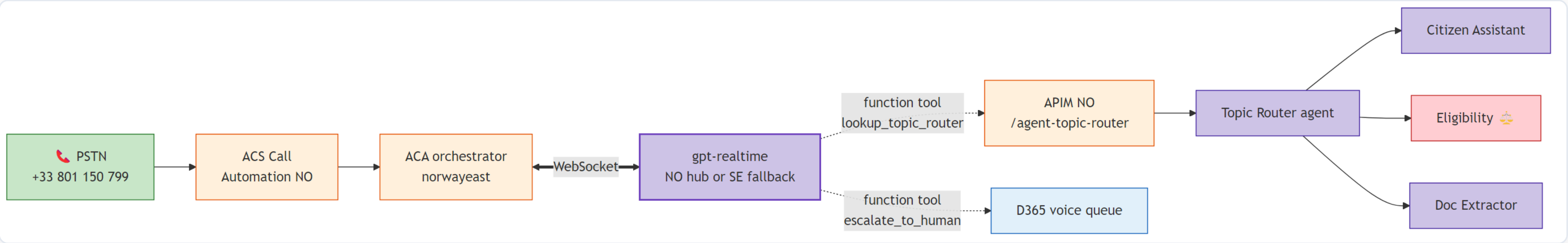
> Six specialist agents, two function tools, one orchestrator: seven agents in all, always under one human caseworker.

Agent catalogue.

Agent	Purpose	Model	EU AI Act class
Topic Router	Orchestration · 12 languages · slot-filling	Fast	Limited
Request Classifier	Intent · agency · language · urgency	Fast	Limited
Translator	12 languages · keeps admin terms exact	Strong + Translator	Limited
Eligibility Pre-Assessor ⚖️	Result + evidence · runs in a protected enclave · ledger-logged	Strong + rules	High (public service)
Citizen Assistant	Answers only from the cited public knowledge base	Strong, grounded	Limited
Document Reader	Passport / payslip / lease extraction	Fast + Doc Intelligence	Limited
Caseworker Helper	Summarise · draft replies · suggest next action	Strong, grounded	Limited

No agent makes a final decision. Every result is a *proposal* to a human caseworker, who confirms, changes or rejects it.

Voice channel: the model uses tools on its own.



Citizen dials → the call lands on Azure Communication Services → a small service opens a two-way audio stream to the speech model. The model decides on its own to call `lookup_topic_router` or `escalate_to_human` (a warm transfer). This is the Microsoft Agent Framework *Agents-as-Tools* pattern, on a real phone call.

Eligibility model lifecycle.

01 **Champion-challenger**

A new version runs in shadow on 5% of traffic for one week, then we evaluate it.

02 **Parity gate per language**

A gold test runs in all 12 languages. If one language is too weak, the release is blocked.

03 **Drift detection**

A daily test on inputs and outputs. Drift opens a security incident and forces a new evaluation.

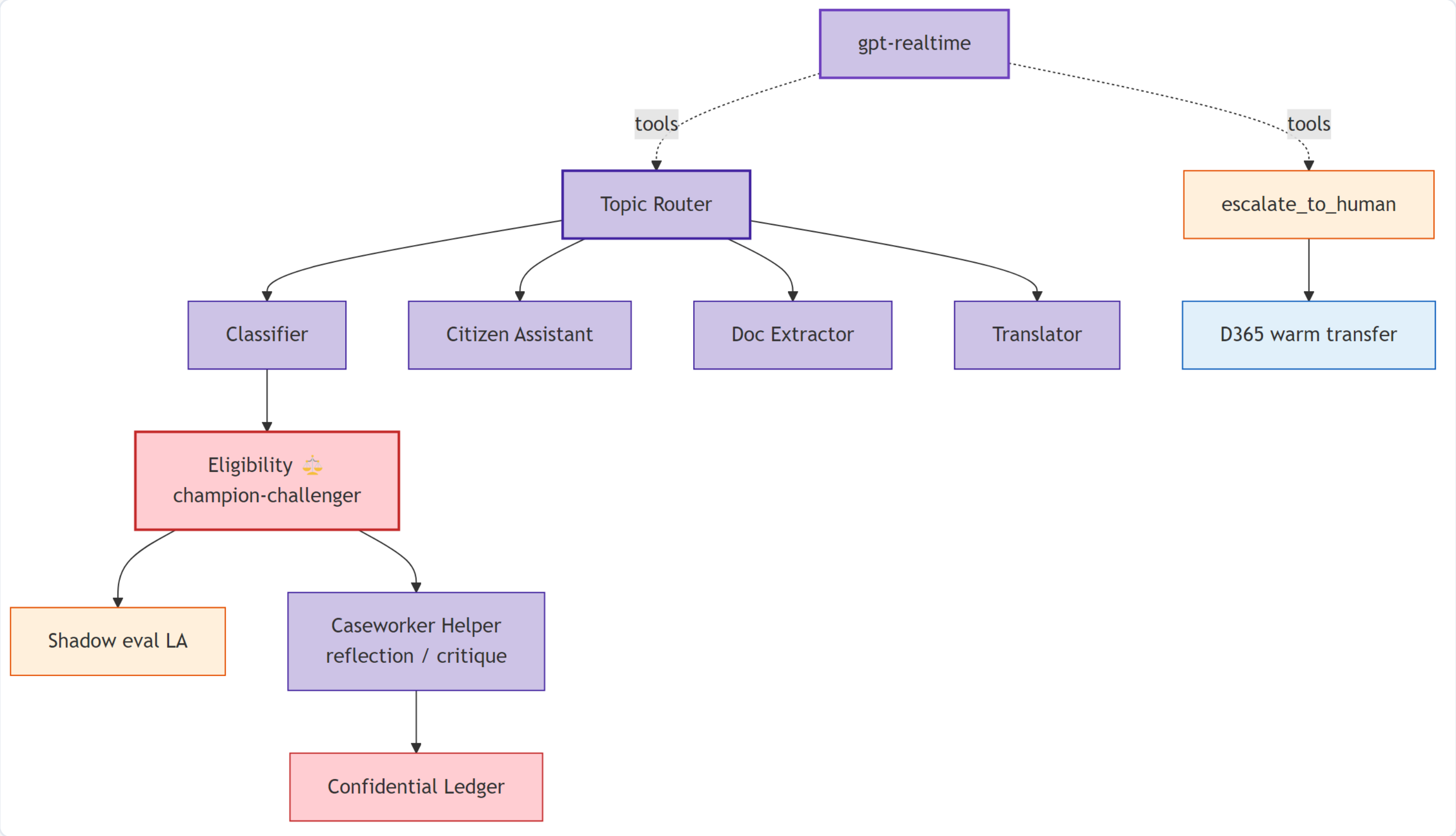
04 **Bias monitoring**

Fairness tests across age, location and channel for 30 days. A problem routes cases to a senior reviewer.

05 **Rollback in seconds**

Versions never change. Switching the alias restores the previous one at once, with an audit record.

Multi-agent coordination: beyond a single chatbot.



Handoff (router → six experts) · Reflection (the helper reviews the eligibility result) · State graph (the case is a 6-state workflow that can undo a step) · Function tool and warm transfer on voice · Shadow / canary for every model change.

Trust built into the brain.

Content Safety
A jailbreak and prompt-injection detector runs on every turn; a block raises a security event.

Evaluations
Quality is tested on every change: language parity, grounding and safety.

AI Act register
Each agent is declared with its purpose, risk level and limits, in a versioned file.

Confidential Ledger
Each high-risk result is hashed into a log that cannot be changed.

Human-in-the-loop
No agent decides. A caseworker confirms, changes or rejects every result.

Transparency
The citizen is always told AI is used: a chat badge and a spoken message in 12 languages.

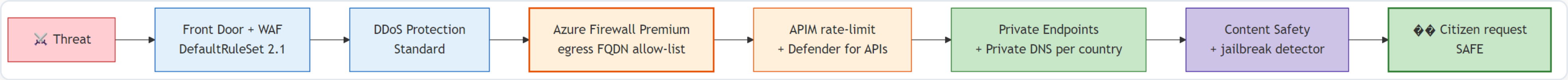
The Eligibility agent is the only high-risk part. It runs inside a protected enclave (its data is encrypted in memory, even from an administrator), and it never decides alone.

Security & compliance.

03

How we made it provable: eight EU laws mapped to controls in code

Defence in depth: many layers.



Edge: application

Azure Front Door and web firewall: common-attack rules, bot protection, rate limits

Edge: network

Standard DDoS protection on every network

Traffic out

Azure Firewall · destination allow-lists · TLS inspection

Identity & data sovereignty.

IDENTITY

- One citizen-identity system per country (Microsoft Entra External ID)
- National eID via a certified broker: MitID · BankID + Freja+ · ID-porten
- Support for the EU Digital Identity Wallet
- Admins get time-limited access only, through one jump-host

DATA: FIVE ZONES, ALL IN-COUNTRY

- **Live**: managed database and cache
- **Documents**: write-once, cannot be changed
- **Conversations**: kept for AI Act records
- **Knowledge**: search, locked per citizen and country
- **Analytics**: sovereign EU Fabric capacity

> One encryption key set per country · modern TLS in transit · two-way TLS to partners · field-level encryption for national IDs · public network access turned off everywhere.

How we made it compliant: the story.

01 We started from the law, not the features

Eight bodies of EU law touch one interaction: a single voice call is data protection, accessibility, e-identity, cybersecurity and AI law, all at once.

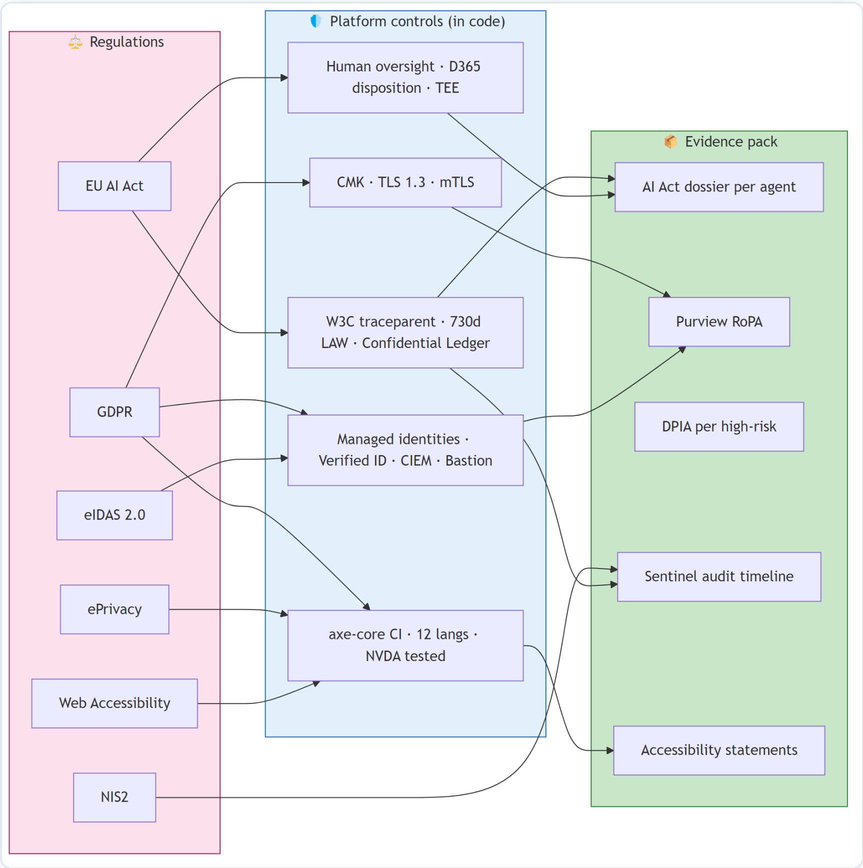
02 Each obligation became a control in code

Every rule maps to a real platform mechanism, with an evidence file behind it.

03 We made the evidence provable

One trace number follows every interaction, from the browser to the model and back, kept for two years.

GDPR: data protection · EU AI Act: high-risk AI · eIDAS: e-identity · NIS2: cybersecurity · ePrivacy: cookies · WCAG: accessibility



EU AI Act: article by article.

Article	Requirement	UDCSP delivery
Art. 12	Automatic records ≥ 6 months for high-risk	Model logs kept 730 days = 2× the minimum
Art. 13	Transparency for deployers	A register entry per agent · tests per language
Art. 14	Human oversight	A caseworker confirms every result · ledger-logged
Annex III §5(b)	High-risk = access to essential public services	The Eligibility agent is declared high-risk
Art. 50	The citizen must know AI is used	AI badge on chat · spoken message in 12 languages

> The AI Act, GDPR, ePrivacy, eIDAS, NIS2 and accessibility law are each mapped to platform controls. The full mapping is in `docs/biz/datacompliance.md`.

Security in action: a prompt injection, stopped.

A message that tries to leak the hidden instructions, or change a result, is caught at three independent layers (three different Azure tools):

01 **Azure API Management**

The gateway sees the strange token rate and blocks the data leak.

02 **Azure AI Content Safety**

The jailbreak detector raises a security event and opens an incident.

03 **Fixed eligibility rule**

A deterministic plug-in rejects the request before any model runs.

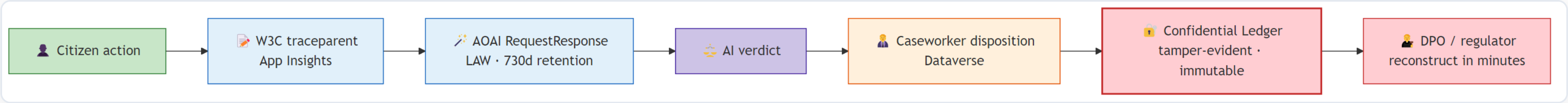
THE AUTOMATIC RESPONSE

- The session is isolated at once
- The citizen flow is brought back safely
- An audit pack is saved for review

0

citizen data exposed

Compliance in action: replay a decision, months later.



Hans, a data-protection officer, takes the citizen's trace number and rebuilds the whole decision: the model, the tokens, the result, the human choice, and the cryptographic ledger proof.

The AI Act minimum is six months. We keep two years. From the request to a full audit pack: under 10 minutes.

Monitoring & FinOps.

04

One anonymized central view · end-to-end tracing · cost per token

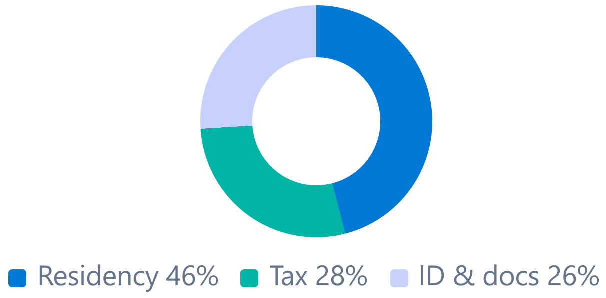
One central view: anonymized, in Fabric & Power BI.

Leadership needs the big picture. But raw citizen data never leaves its country. So only anonymized, aggregated numbers travel to a central view.

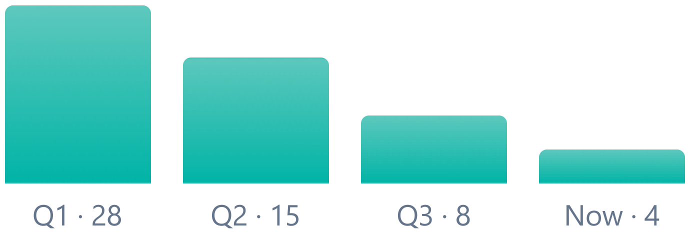
12,480	4 days	71%	4.5 / 5
cases this month	median time (was 28)	AI-assisted, human-approved	citizen satisfaction

CASES BY COUNTRY (ANONYMIZED)

BY SERVICE



MEDIAN DAYS TO DECISION



Only anonymized, aggregated numbers leave each country: never a name, never a case file. Built on Microsoft Fabric and Power BI · refreshed every hour.

FinOps: cost under control.

ALLOCATION & SHOWBACK

- Every resource tagged `country`, `workload`, `cost-center`
- Cost views per country and per workload
- Each ministry sees its own country's cost

TOKEN BUDGET & CAPACITY

- A monthly token budget per agent, written in code
- The build fails if the total is over the pool
- Reserved capacity for steady models · pay-as-you-go for peaks
- Alert on an unusual daily cost jump

> We treat AI tokens like any other budget line: planned, tagged, and checked at build time.

Cost & economics.

05

What it runs · what it costs per citizen · and how we keep it low

Three tiers, one platform

€60k
/ month · pilot · 30k citizens

€144k
/ month · regional · 300k

€5.28M
/ month · national · 22.3M ceiling

€2.84
per citizen / year at scale

Scale	Citizens	Active / mo	Peak sessions	Run-rate / mo	Per citizen / yr
Pilot · early band	30 000	12 000	≈ 240	≈ €60 k	≈ €24
Regional · early band	300 000	120 000	≈ 2 400	≈ €144 k	≈ €5.8
National · DK + SE + NO	22 300 000	8.92 M	≈ 178 000	≈ €5.28 M	≈ €2.84

> The same platform sized for three population bands. National = real populations, full model in `docs/biz/cost.md`.

Same platform, three regions, and the region changes the price

Country	Pop.	Azure region	Index	€ / cit. / yr
🇩🇰 Denmark	6.0 M	North Europe	1.00	€2.69
🇸🇪 Sweden	10.7 M	Sweden Central	1.08	€2.83
🇳🇴 Norway	5.6 M	Norway East	1.20	€3.04
All	22.3 M	three regions	-	€2.84

Index = Azure-infrastructure price vs North Europe. Licences (D365, identity) are region-neutral.

Norway East premium + **smallest population** → highest per-citizen cost. No country subsidises another.

> Per citizen / year: DK €2.69 · SE €2.83 · NO €3.04. The region, not the code, drives the gap.

Cost over time: adoption ramps, so the bill ramps

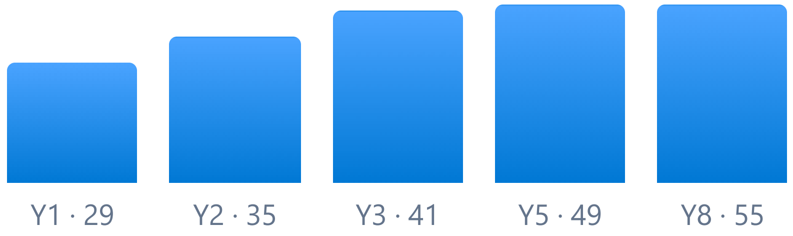
×5.3
citizens, Y1 → Y8

×1.9
yearly bill grows

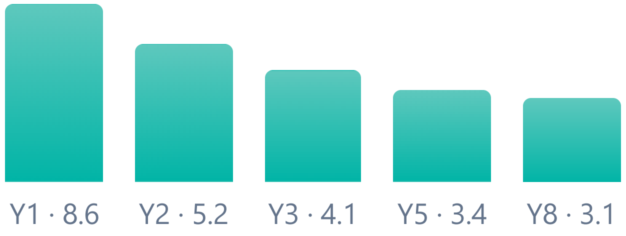
€8.6 → €3.1
per citizen / year

≈ €1.9M
fixed floor / month

RUN-RATE / YEAR (€M): GROWS SLOWLY



PER CITIZEN / YEAR (€): FALLS AS IT MATURES



ADOPTION

Y1 **15%** · 3.3M

Y3 **45%** · 10.0M

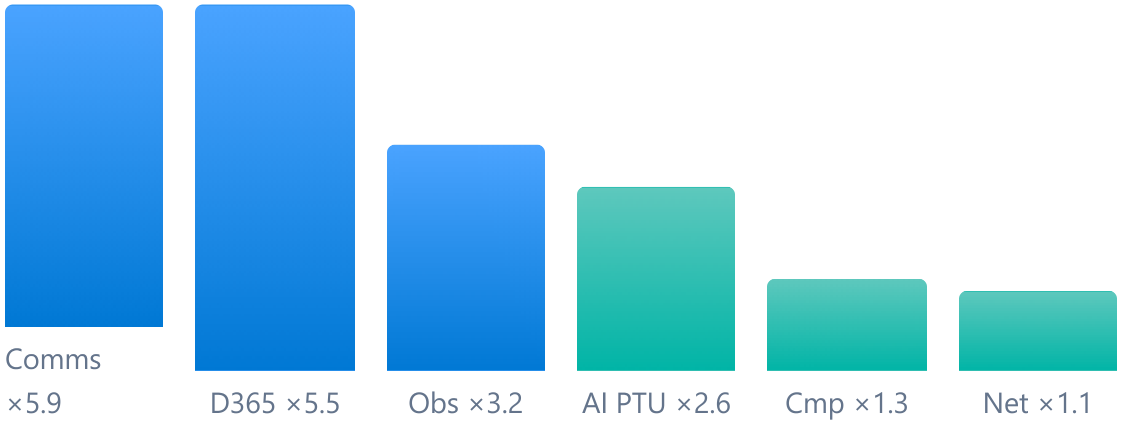
Y5 **65%** · 14.5M

Y8 **80%** · 17.8M

Population fixed at 22.3M. More adoption = cheaper per head.

> Citizens ×5.3, bill only ×1.9: the fixed floor is paid once, then amortised. Per-citizen falls €8.6 → €3.1.

Not every service scales the same way



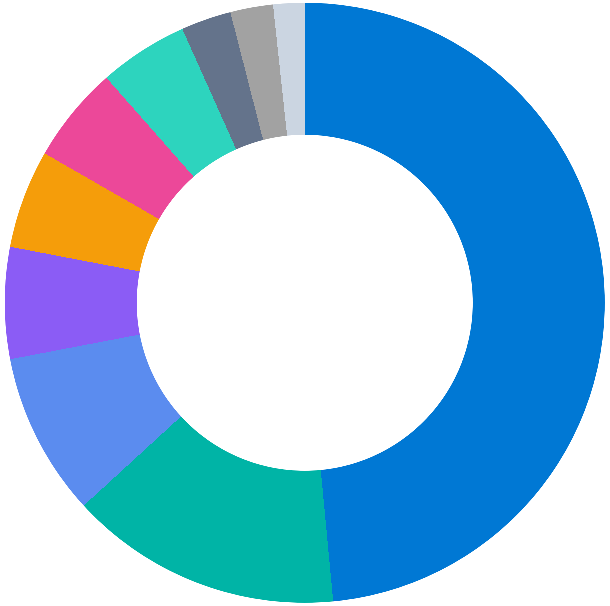
Cost growth Y1 → ceiling, while active users grow **×6.7**

Service	Shape	Y1→ceiling
ACS SMS · voice · email	📈 metered	×5.9
Dynamics 365 caseworkers	👤 headcount	×5.5
AI PTU pools	🧱 block	×2.6
APIM · Logic Apps · SKUs	🪜 stepped	×1.3
Firewalls · Front Door · net	⬛ flat	×1.1

APIM & **Logic Apps** jump in SKU steps, not per user. **AI PTU** is reserved to peak; the mini-first router keeps it flat.

> Only ≈ €3.4M of the €5.28M ceiling scales with citizens; ≈ €1.9M is built once. +1 active citizen = €0.38 / month, €0 floor. Full breakdown in `docs/biz/cost.md` §7.

Where the money goes: €5.28M / month at the ceiling



- **AI & Foundry**: €2.56M · 48%
- **Dynamics 365** (caseworkers): €780k · 15%
- Network & Security: €465k · 9%
- Observability: €317k · 6%
- Data & Caching: €281k · 5%
- Compute & Integration: €273k · 5%
- Communications: €255k · 5%
- Identity · Analytics · Governance: €352k · 7%

Top two groups ≈ **63%** of €5.28M / month

> Every slice traces to a deployed service; per-group arithmetic in `docs/biz/cost.md` §3-5 & §11.

Fixed floor vs variable: the cost flips with scale

Pilot · €60k / month, mostly fixed floor

National · €5.28M / month, mostly real usage

Driver · per active citizen / month	Value	What it hits
AI conversations (≈ 6 turns)	0.8	reserved AI capacity
SMS	0.6	Communications
Eligibility assessments	0.15	strong model + enclave
Escalation to a caseworker	≈ 4%	Dynamics 365 licences

> Fixed = three sovereign zones · Variable tracks active citizens. Peak concurrency ≈ 2% of active users; that is all we reserve.

Keeping it low: levers already built in

LIST → COMMITTED

−30% to −40% off the AI & compute lines: same citizen experience.

Lever	What it does	Saving
Reservations & Savings Plans	1-3 yr commit on PTU, App Service, PostgreSQL, Fabric	−30 to −40%
Model routing (mini-first)	cheap mini routes; strong model only to reason	≈ 10× / token
PTU right-sized to peak	reserve to peak concurrency · pay-as-you-go spikes	no off-peak waste
Push & email before SMS	SMS is the most usage-sensitive line	can halve comms
Telemetry sampling	adaptive sampling · basic logs · cheap archive	−40 to −60% obs
Tagged showback	build fails if an agent goes over budget	stops drift early

> List ≈ €63.4 M/yr national → ≈ €40-46 M committed. Full lever list in `docs/biz/cost.md` §9.

Performance & reliability.

06

Fast for the citizen, always available: measured targets, automatic failover, chaos drills

Performance & reliability.

99.9%
citizen web target, per country

≤ 2 s
voice answer, 95% of calls

≤ 1 h
recovery point: case data (15 min for drafts)

4 h
recovery time to the backup EU region

Active-passive per country

Azure Front Door sends traffic to the healthy region. Failover in under 5 minutes.

Backup & disaster recovery

Database, cache, storage and agent hosts. One recovery vault per country.

Chaos drills

We break things on purpose: region failover, network cut, database failover. Monthly in test, quarterly in production.

Autoscale & caching

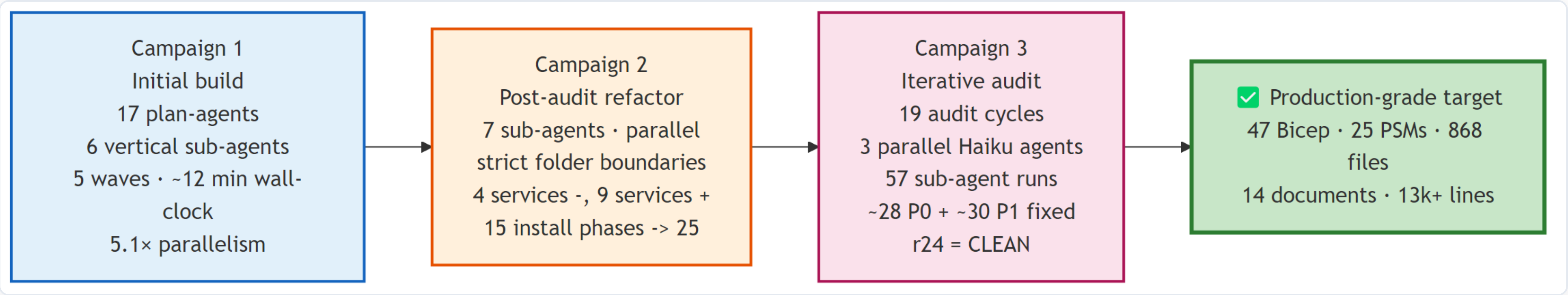
Services scale out under load. Reads are cached at the gateway, so peaks stay fast.

07

How we built it.

The platform itself was built by an agent swarm, in under 45 minutes of wall-clock time

UDCSP was built by an agent swarm.



3
multi-agent campaigns

~5×
faster than sequential

24
audit rounds to first clean

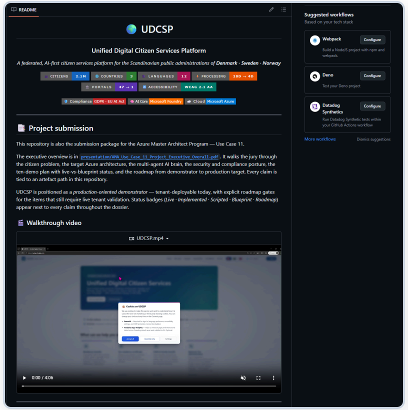
~1,000
files tracked in the repo today

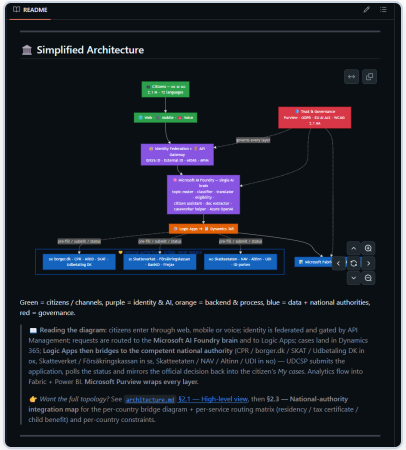
> Three campaigns (build, refactor, then repeated audits) produced the platform. Six sub-agents owned separate folders and ran in parallel. Each round produced fix commits until round 24 was the first fully clean one. The one-command installer is the executable summary of it all.

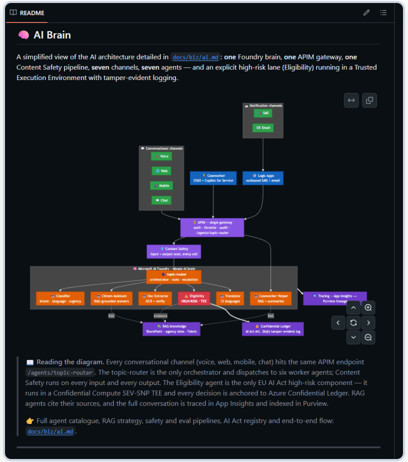
Everything is in the open: one repository.

THE STORY STARTS IN THE DOCS

- `/docs/biz` : the business story: who we build for, every channel, data-and-compliance, demo scenarios, and an acceptance recipe for "done".
- `/docs/tech` : the engineering: architecture deep-dive, data model, network design, one-command install guide, disaster-recovery runbook.
- `/infra` : the Bicep that builds every zone · `/scripts` : the installer.







SHORT LINK
aka.ms/UDCSP

GITHUB
github.com/fredgis/UDCSP

One idempotent deploy: provision.

One script, one click, and you can run it again safely.

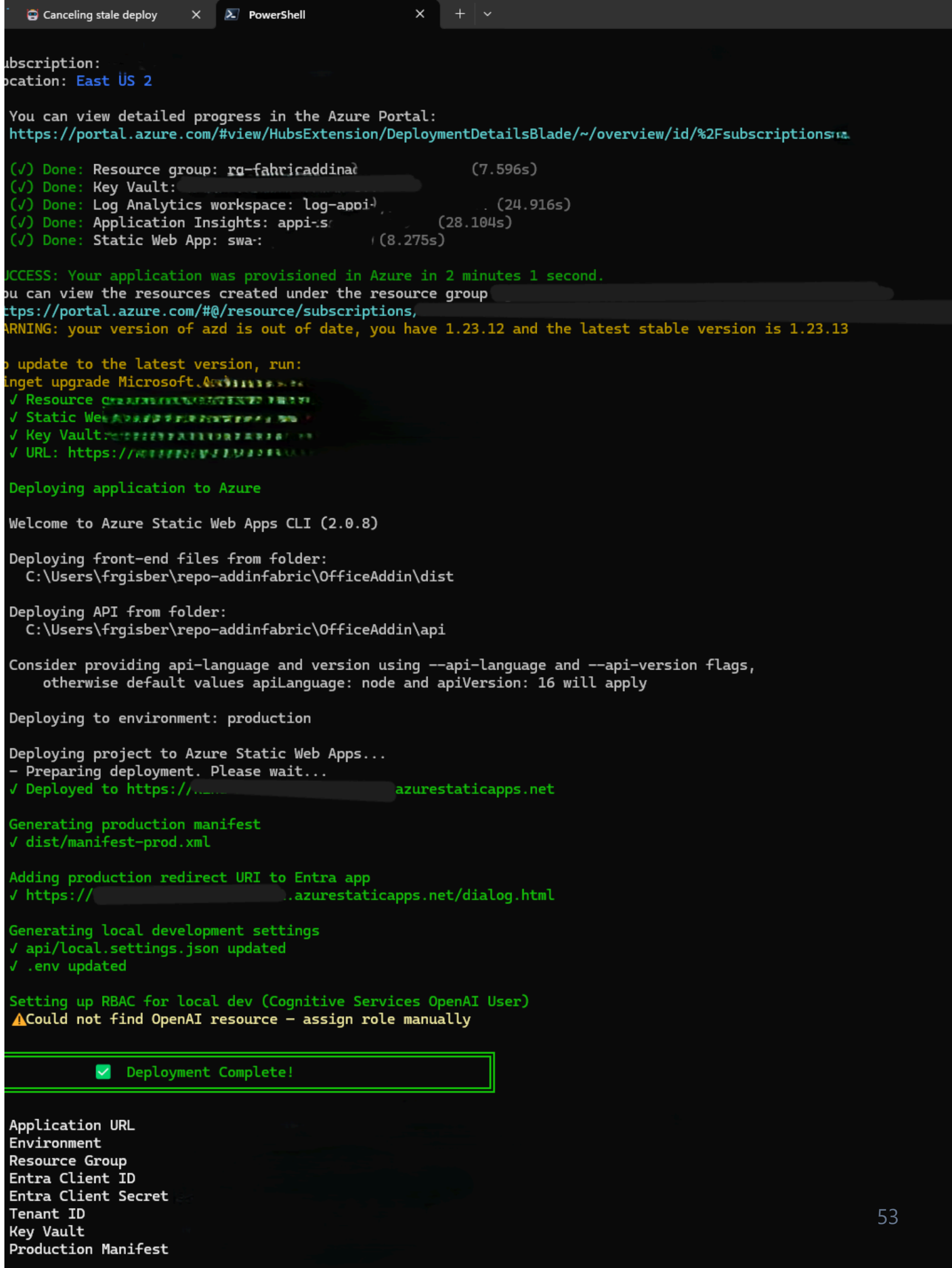
- Checks prerequisites, signs in to Azure, registers the Entra ID identity app.
- Installs, builds, and runs the tests before anything is deployed.
- Provisions all the infrastructure as code via Bicep, through `azd`.
- **Idempotent**: a second run reuses what exists, never a duplicate.

```
Fabric Storyboard Copilot — One-Click Deploy

▸ Checking prerequisites
  ✓ Node.js v24.14.0
  ✓ npm v11.9.0
  ✓ Azure CLI v
  ✓ Azure Developer CLI (azd)

▸ Authenticating with Azure
  ✓ Signed in as
  ✓ Subscription: [redacted]
  Authenticating azd (fresh login)...
  ✓ azd authenticated

▸ Gathering configuration
  Environment name (e.g., dev, staging, prod) [dev]:
  ✓ Environment: dev
  Azure region [eastus2]:
  ✓ Location: eastus2
  Deploy a NEW Azure OpenAI resource? (y/n) [n]:
  Existing Azure OpenAI endpoint URL: https://[redacted].openai.azure.com/openai/v1/
  ✓ OpenAI endpoint: https://[redacted]-openai.azure.com/openai/v1/
  ✓ OpenAI deployment: gpt-4o

▸ Configuring Entra ID
  Create a NEW Entra ID app registration? (y/n) [y]:
  Registering app: Fabric Storyboard Copilot (dev)
  ✓ App registered:
  Setting Application ID URI...
  Exposing access_as_user scope...
  ✓ Scope: api://[redacted]/access_as_user
  Adding redirect URIs...
  Adding API permissions...
  Authorizing Office clients...
  ✓ Office clients authorized (SSO)
  Generating client secret...
  ✓ Client secret created (valid 1 year)
  Creating service principal...
  Granting admin consent...
  ✓ Entra ID setup complete

▸ Installing dependencies
  ✓ Frontend packages installed
  ✓ Backend packages installed

▸ Building application
  ✓ Frontend → dist/
  ✓ Backend → api/dist/

▸ Running tests
  ✓ Frontend tests passed
  ✓ Backend tests passed

▸ Provisioning Azure infrastructure (Bicep via azd)
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

To update to the latest version, run:
winget upgrade Microsoft.Azd
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

To update to the latest version, run:
winget upgrade Microsoft.Azd
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13
```

One idempotent deploy: release and verify.

It deploys, verifies, and reports, every time the same way.

- Provisioned in ~2 minutes, then the app is deployed to Azure.
- Generates the production manifest, wires redirect URIs and RBAC (role-based access control).
- Ends with Deployment Complete and a summary of every resource and URL.
- The same script runs in the pipeline on every change, repeatable, not hand-made.



Real scenario.

Life imitates the demo: a story we did not plan

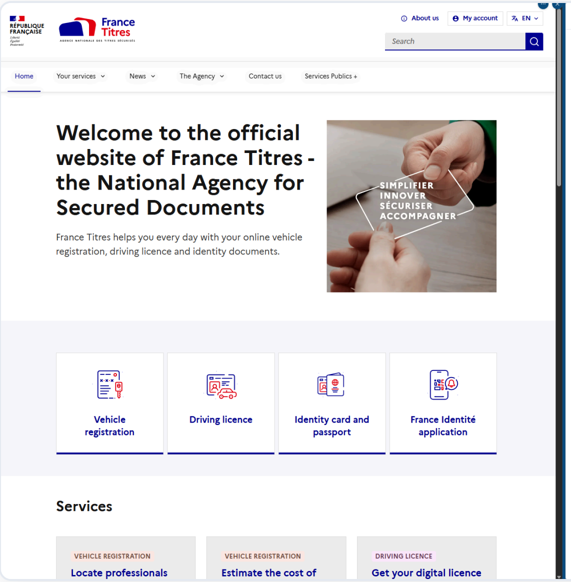
08

"I bought my car back." 🚗

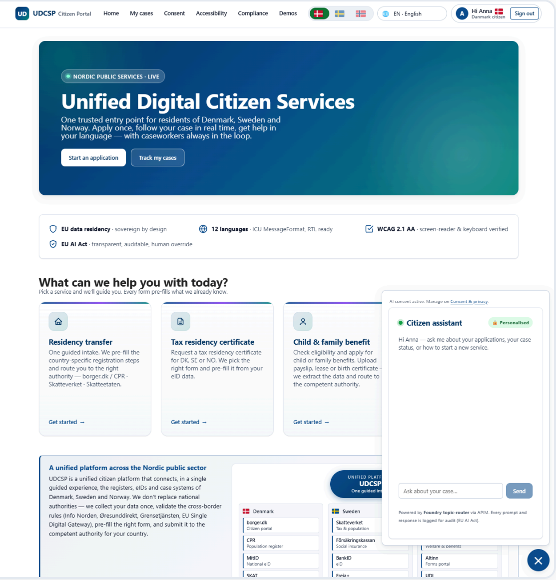
To change my licence plate, I signed in to the official French government portal: France Titres.

I had never seen it while building UDCSP.

And there it was: the same one-front-door layout, the same service cards (vehicle registration, driving licence, ID & passport), the same assistant in the corner. 😄

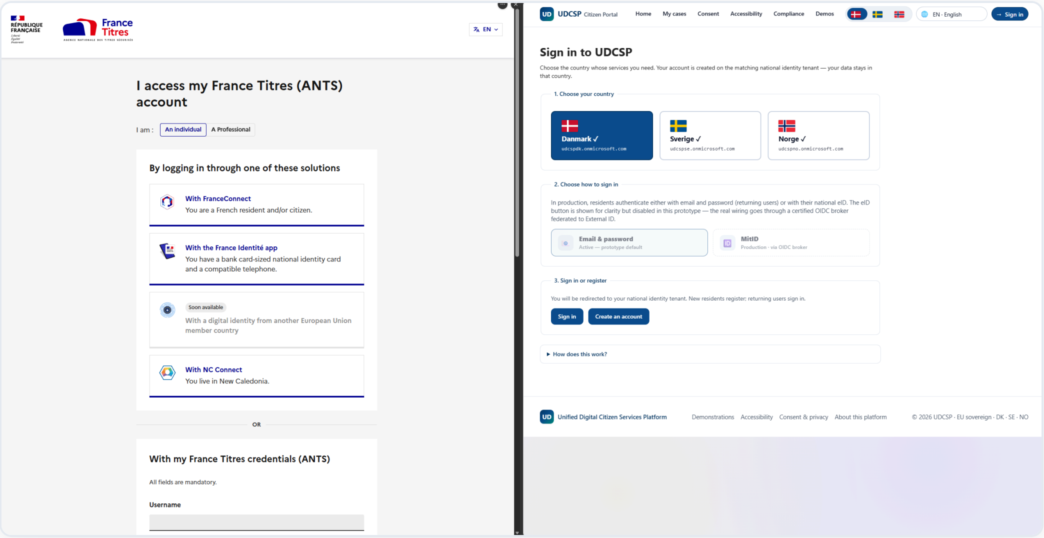


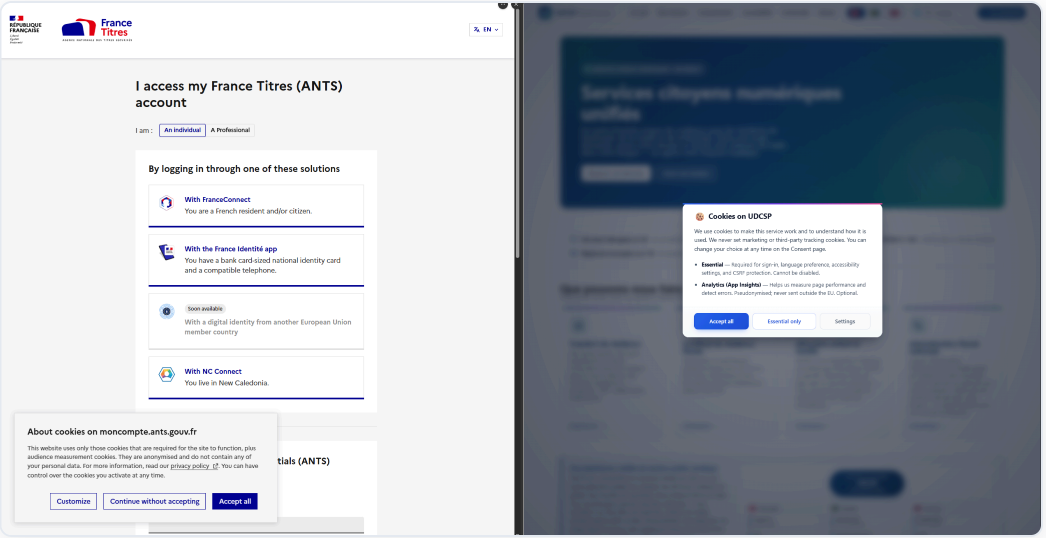
FR FRANCE TITRES, THE REAL THING



UDCSP, OUR DESIGN

...and we'd built it first. 🙂





> Pick your country, sign in with a national eID, and ("*coming soon*") an EU cross-border identity. That is our model, almost line for line. We designed it before we ever saw it. 🙂

Status & roadmap.

09

What runs today · what's blueprint · how it reaches production

Requirement coverage: the honest scorecard.

Unified identity
Entra External ID · eIDAS-ready
Live · Verified ID

12-language portal
WCAG 2.1 AA accessibility
Implemented

AI assistant
web · mobile · voice
Live

Cross-border case
DK → SE pre-fill + saga
Live · saga

Eligibility + human-in-loop
verdict, then a person decides
Implemented · live TEE

EU AI Act evidence
article-by-article trail
Implemented · live ledger

GDPR rights
records · access · erasure
Scripted

Sovereign monitoring
9 workbooks, per country
Live · Fabric F64

One-command deploy
25 phases, idempotent
Live

Live: today · Implemented: merged & tested · Scripted: idempotent installer · Blueprint: designed & gated

The working core carries the demo; each blueprint brick has a named gate.

Four gates from accelerator to production.

GATE 1 · WK 1-2

Tenant validation

Confirm model deployments, Fabric F64 capacity, DDoS plan & Foundry hub.

GATE 2 · WK 3-6

Live confidential compute

Eligibility on a SEV-SNP enclave + Confidential Ledger. **Demo 6 → Live.**

GATE 3 · WK 6-14

Partner-agency integration

Mutual-TLS agreements + production federation hub. **Demo 1 → real authority.**

GATE 4 · WK 14-20

D365 + Copilot for Service

Strangler-fig repoint of case writes. **Demo 5 → Live.**

Today → Gate 1 → Gate 2 → Gate 3 → Gate 4 → production. Each gate is independently shippable: every blueprint brick has an owner, a window, and a demo it flips to live.

UDCSP Guardian.

The next era: proactive, autonomous, human-supervised AI

10

From reactive to proactive.

20 to 60%

of eligible citizens never claim the benefits they are entitled to (OECD). Not fraud, not choice: they do not know.

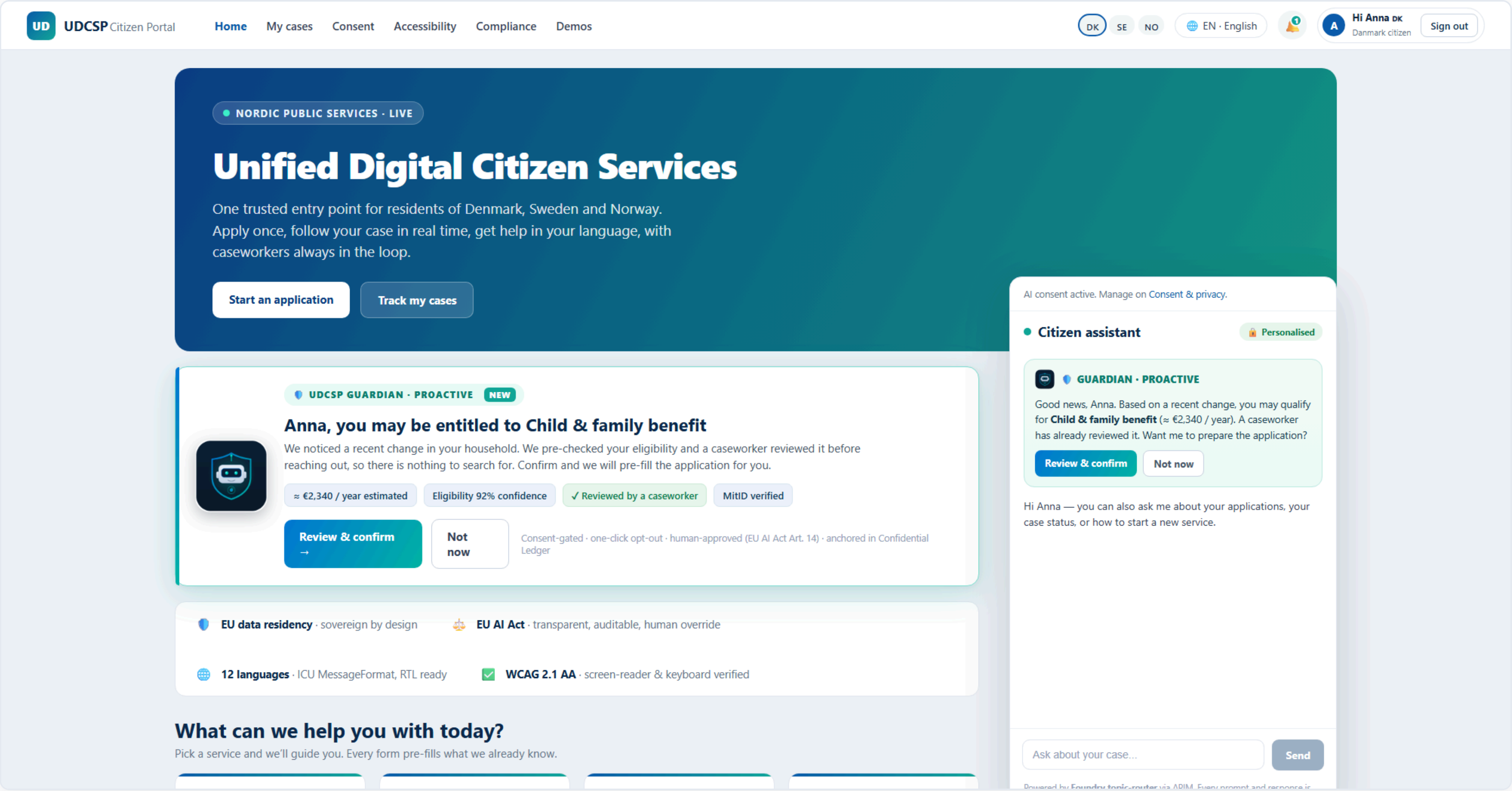
Today UDCSP is reactive: the citizen must know they qualify, find the portal, and apply. Guardian is the step after the EU once-only principle: no-stop-shop.

WHAT CHANGES

- The state detects the life event, not the citizen
- It reaches out first, in the citizen's language
- A human caseworker approves every message before it is sent
- Every step is consent-gated and ledger-anchored

> Guardian turns a faster front door into a state that reaches out to the people it is meant to serve.

How Guardian appears to Anna.



A proactive, caseworker-approved entitlement, surfaced right in the portal Anna already uses.

How Guardian works: one autonomous loop.

01 **Event Scanner detects a life event**

An autonomous planner scans in-country data: a birth, a move, age 67, an income drop.

02 **Eligibility runs in shadow**

The existing high-risk agent assesses silently, rule by rule, with no application attached.

03 **Caseworker Helper drafts the outreach**

A short, cited message in the citizen's language, from the next-best-action catalogue.

04 **Critic agent reviews it** reflection

A new agent checks legal basis, tone and false positives, and can send it back.

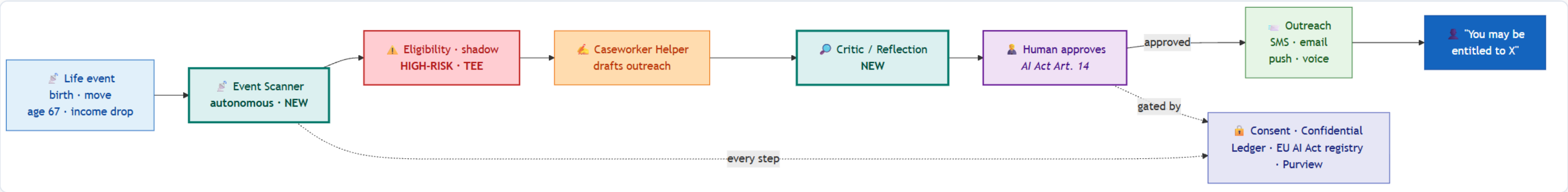
05 **A human approves, adjusts or rejects** AI Act Art. 14

The loop cannot advance without a caseworker. Nothing autonomous past this gate.

06 **Outreach goes out, every step anchored**

SMS, email, push or voice, consent-gated, hashed into Confidential Ledger.

The architecture: new engine, reused brain.



MOSTLY ASSEMBLY

- reused Eligibility (shadow), Caseworker Helper, Translator, the outreach channels, the ledger and registry
- new Event Scanner, Critic agent, one outreach workflow, one KPI tile

SOVEREIGN AND SAFE

- A Danish signal is assessed by the Danish brain, no border crossing without consent
- The high-risk lane is unchanged: sealed enclave, ledger-anchored, never decides

> Two new agents wrapped around a platform that is already live, private and governed.

Responsible autonomy, and the number that matters.

RESPONSIBLE BY DESIGN

- **GDPR Art. 22** · never an automated decision with legal effect; a human approves
- **EU AI Act Art. 14** · mandatory human oversight kept in the loop
- **Consent** · lawful basis required, one-click opt-out, checked before sending
- **Ledger** · signal, verdict, draft and disposition all anchored

€ recovered

the new executive KPI: unclaimed entitlements proactively delivered, by country and language

FROM EFFICIENCY TO EQUITY

- Reactive story: 28 days to 4
- Guardian story: money and rights reaching people who never knew

> Proactive government is safe when autonomy stops at a human, the basis is lawful, the citizen can opt out, and every step is provable.

Demo.

10 minutes, live on a real tenant

11

The run of show.

01 Anna moves DK → SE · Live

Sign in with a Danish eID · upload a passport and lease · the AI reads, translates and proposes a result · consent · the case crosses the border.

02 Lars calls the voice line ⭐ · Live

A blind citizen dials a real free number · the model answers in Norwegian and routes itself · a human transfer is offered.

03 Maria uses a screen reader in Polish · Live

The whole portal in Polish, keyboard only, screen-reader clean: accessibility is a citizen right.

04 Erik photographs a payslip on iPhone · Live

The same web app on mobile · native iOS picker · structured fields and a result, inline.

05 Ole builds it from a clean tenant · Live

One command: 25 phases, sample data seeded, smoke tests green.

> Hero moment: a real phone call: dial `+33 801 150 799`, hear the model answer, and watch the live dashboard update within two minutes.

One command, from clean tenant to running platform.

```
git clone https://github.com/fredgis/UDCSP
cd UDCSP
pwsh ./scripts/install/Install-UDCSP.ps1 `
     -Environment dev -Zone all `
     -SeedSyntheticData -Verbose
```

- 25 phases in order: landing zone → identity → security → data → monitoring → gateway → Foundry → Dataverse → frontend → voice → governance → quality
- Sample data is seeded in parallel with the frontend
- A smoke test runs at the end (identity · gateway · AI · case creation · dashboard · accessibility)
- An HTML report is saved in `scripts/install/reports/<timestamp>/`

> From `git clone` to a working federated platform with real sample data: one command. The same script runs in our pipeline on every pull request.

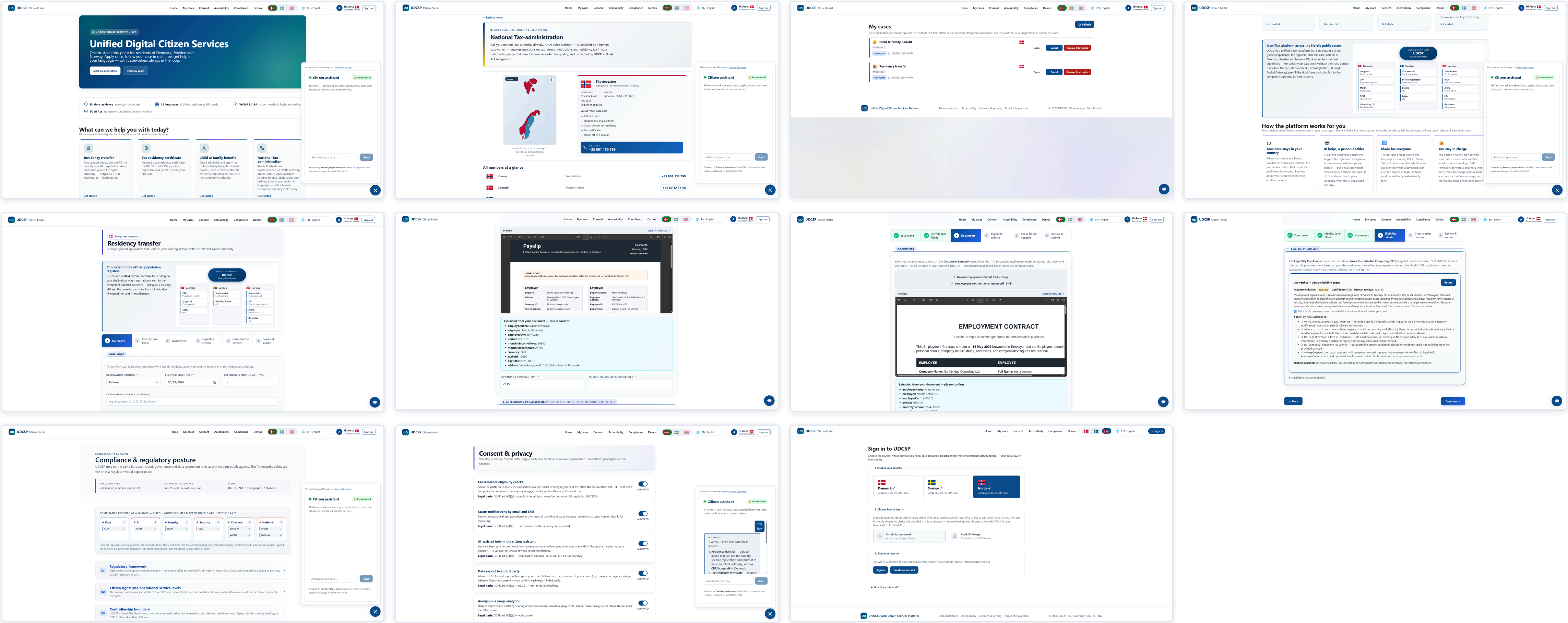
CLOSING

UDCSP: a production accelerator with a named path to production.

3 sovereign zones · 7 AI agents · one private path from the portal to the case · every decision anchored to a regulation · every claim labelled and provable on a real Azure tenant. A working core you deploy with one command today, and a four-gate roadmap that takes the blueprint bricks to production. A real French government portal, seen by accident, proved the architecture was right.

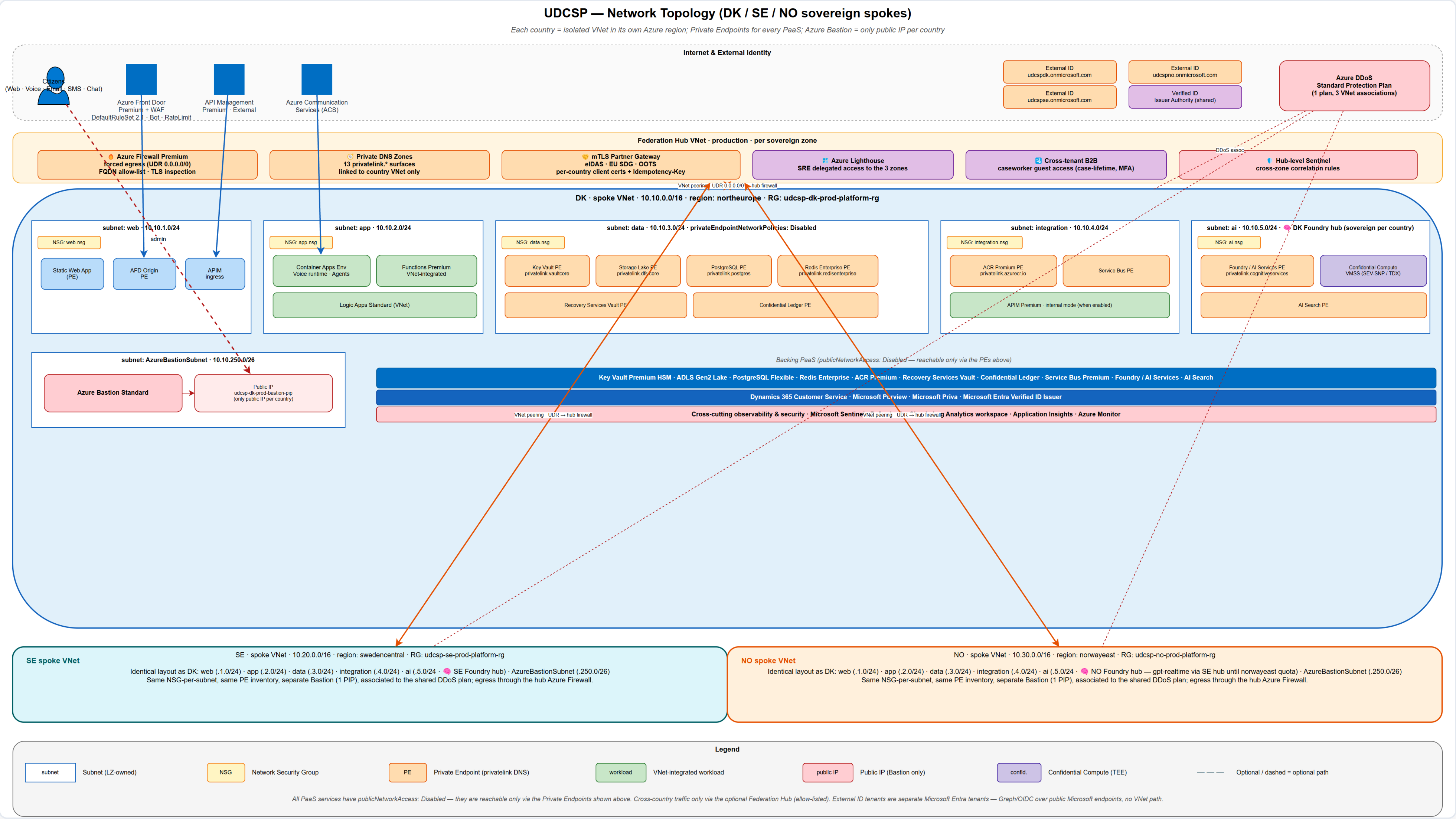
Frederic Gisbert · August 2026 · github.com/fredgis/UDCSP

Annex: the platform, screen by screen.

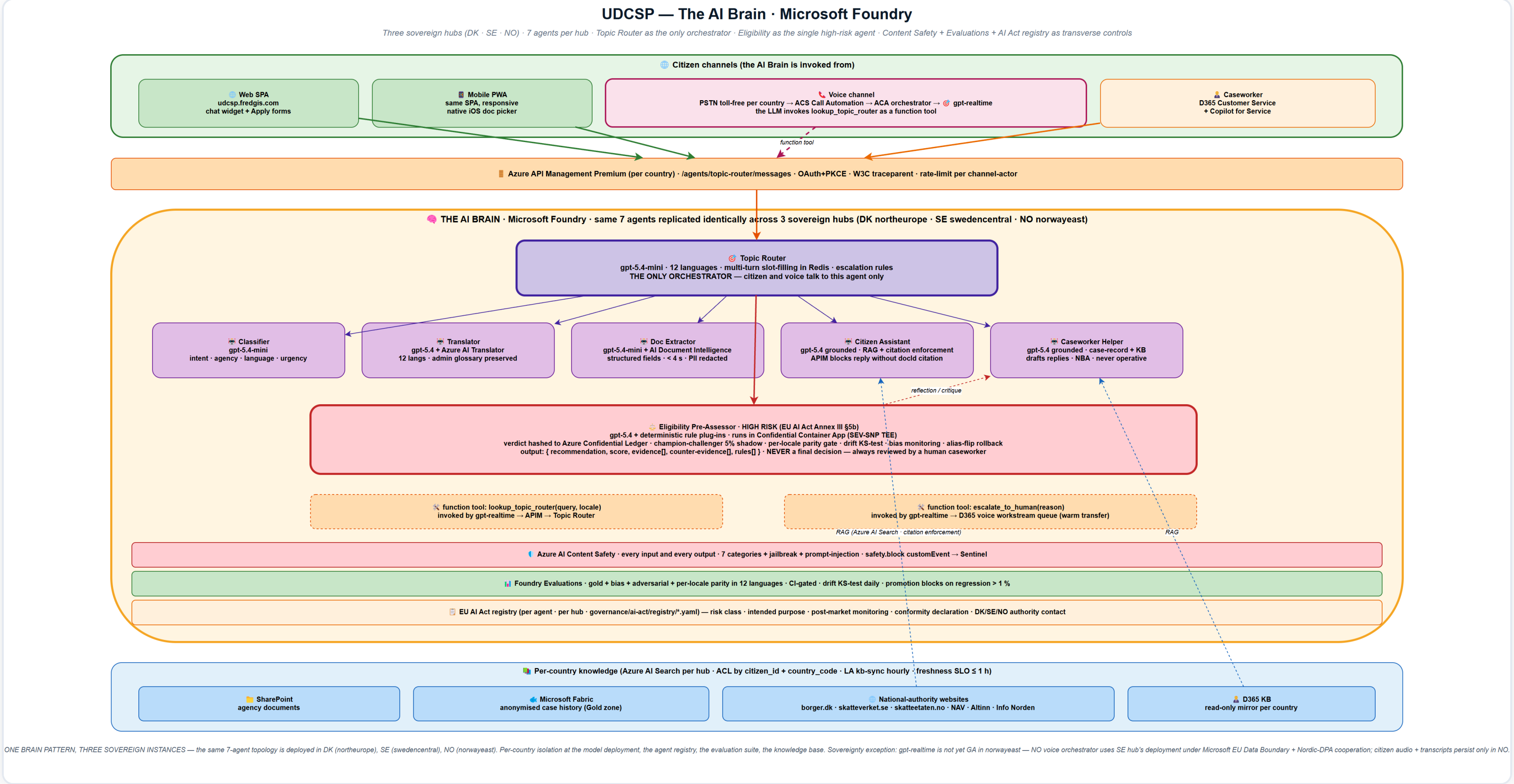


1 Home · 2 Contact / voice · 3 My cases · 4 Assistant · 5 Residency intake · 6-7 Document reading · 8 Cross-border result · 9 Compliance · 10 Consent · 11 Sign-in. One web app, twelve languages, three channels, using the same React and TypeScript code on desktop, tablet and phone.

Annex: network topology in full.



Annex: the AI Brain blueprint in full.



One orchestrator and six specialists on Azure AI Foundry, the high-risk eligibility agent in a sealed enclave, and the EU AI Act register wired into every model call: the dense source map behind the simplified gateway slide.

Annex: telemetry pillars in full.

METRICS / LOGS
Azure Monitor
3 workspaces, one per country · 180 days hot · 7 years cold · never joined

DISTRIBUTED TRACE
Application Insights
One trace number end-to-end · front door → gateway → workflow → Dataverse → agent → model

AI QUALITY
Foundry Evaluations
Continuous · drift monitors · language parity · bias monitoring

AI ACT EVIDENCE
Confidential Ledger
Records that cannot be changed · joins logs, AI and Dataverse in one query

ACTIVE
Synthetic + real-user
5 external regions · 60-second probes · load times per language

DASHBOARDS
9 workbooks
platform health · citizen-journey funnel · AI-decision traces · per country